

Dirichlet realizations, spectral projectors, and local extensions

The central construction in this lesson is a bounded point-evaluation functional on a spectral subspace. Elliptic estimates bound that functional, and its Hilbert-space representative produces a spectral kernel. This construction works for a sharp spectral cutoff even when the spectrum is continuous. Compactness has a separate role: on a bounded smooth Dirichlet domain it supplies an eigenbasis and a finite trace.

Throughout, $D_j = -i\partial_j$, and Hilbert inner products are linear in the first variable. The precise analytic entry results used below are the integer half-space trace and extension construction in Section 2 of Boundary energy, local inverses, and harmonic data, supplemented by the complete real-order trace proof in Section 2 below; the local boundary H^1 -to- H^2 theorem Section 6 of Boundary energy, local inverses, and harmonic data, with Lipschitz principal and bounded lower coefficients; its smooth integer bootstrap Section 7 of Boundary energy, local inverses, and harmonic data; and the interior counterpart Section 8 of Local inverses and distance-weighted elliptic estimates together with the differentiated-equation argument. We check their hypotheses at each use. The other Fourier entry facts are inversion and Plancherel with inverse factor $(2\pi)^{-n}$. For the arbitrary lower-bounded extension in Section 10, the complete spectral construction is proved in Spectral measures with the original operator domain retained, Sections 1–4 of Spectral measures with the original operator domain retained: scalar measures, the bounded positive calculus, all multiplier domains, and the exact inverse correspondence back to the given operator. The bounded-domain calculus is also proved directly in Section 5. With the original inner-product convention and the spectral measure F_A , these proofs give

$$\|m(A)f\|^2 = \int_{\mathbb{R}} |m(t)|^2 d(F_A(t)f, f), \quad D(A^q) = \left\{ f : \int |t|^{2q} d(F_A(t)f, f) < \infty \right\}. \quad (\text{DSP1})$$

The analytic entry results are used with their stated hypotheses. The spectral identities just displayed have complete proofs in the companion, including the recursive domains of the original operator. The elliptic regularity-to-kernel implication and the boundary-domain implication are proved here. Nothing below identifies this second-order scalar result with a theorem for general systems or higher-order boundary conditions.

1. Weighted Dirichlet realization

Let $n \geq 1$, and let $X \subset \mathbb{R}^n$ be a nonempty bounded open set with smooth boundary. Let $r \in C^\infty(\overline{X})$ be real and strictly positive. Consider the unchanged differential expression

$$Pu = -\partial_j(g^{jk}\partial_k u) + b^j\partial_j u + cu, \quad (u, v)_r = \int_X u\bar{v}r \, dx. \quad (\text{DSP2})$$

Repeated indices are summed from 1 to n . The coefficients are smooth on a neighborhood of the closure; $g^{jk} = g^{kj}$ are real and $g^{jk}(x)\xi_j\xi_k \geq \theta|\xi|^2$ there for some $\theta > 0$. The coefficients b^j, c may be complex. Assume exactly that P is symmetric on $C_c^\infty(X)$ for $(\cdot, \cdot)_r$.

Exact correspondence for an arbitrary real principal matrix. The symmetric presentation above also applies to a given real matrix $G = (G^{jk})$ whose quadratic form is positive, without changing its original coefficients or differential expression. Retain those original data (G, B, c, r) , and define the additional coefficient fields

$$S^{jk} = \frac{G^{jk} + G^{kj}}{2}, \quad A^{jk} = \frac{G^{jk} - G^{kj}}{2}, \quad \beta^k = B^k - \sum_j \partial_j A^{jk}. \quad (\text{DSP2a})$$

For every distribution u , commuting weak derivatives and pairing the (j, k) and (k, j) terms give

$$\sum_{j,k} A^{jk} \partial_j \partial_k u = 0, \quad -\partial_j (G^{jk} \partial_k u) + B^k \partial_k u + cu = -\partial_j (S^{jk} \partial_k u) + \beta^k \partial_k u + cu. \quad (\text{DSP2b})$$

The product rule with smooth coefficients proves the displayed expansion for distributions as well as for smooth functions. Also $S^{jk} \xi_j \xi_k = G^{jk} \xi_j \xi_k$ for every real ξ , with exactly the same ellipticity lower bound. Thus the map $(G, B, c, r) \mapsto (S, B - \operatorname{div} A, c, r)$ is an explicit map of coefficient presentations of the same distributional operator on the same weighted space. The condition defining (DSP3) and its operator action are identical for the two presentations. Their adjoints, symmetry, self-adjoint realizations, recursive powers, and spectral projections therefore agree as operators; this follows from equality of their graphs, not from a change of coordinates or of density. Coefficient estimates can be pulled back to the given fields through (DSP2a): for every multiindex α ,

$$\partial^\alpha S^{jk} = \frac{1}{2}(\partial^\alpha G^{jk} + \partial^\alpha G^{kj}), \quad \partial^\alpha \beta^k = \partial^\alpha B^k - \frac{1}{2} \sum_j \partial^{\alpha+e_j} (G^{jk} - G^{kj}). \quad (\text{DSP2c})$$

The triangle inequality gives their exact componentwise norm bounds, including the additional derivative of the antisymmetric part. The original fields remain available throughout this correspondence. Every subsequent formula in the symmetric presentation uses $g = S, b = \beta$ when it is applied to these more general input data.

Define

$$D(P_D) = \{u \in H_0^1(X) : Pu \in L^2(X, r dx)\}, \quad P_D u = Pu. \quad (\text{DSP3})$$

There is a real number $\mu \geq 1$ such that $T = P_D + \mu I$ is self-adjoint, $T \geq I$, and

$$(Tu, u)_r \geq a \|u\|_{H^1(X)}^2, \quad \|u\|_{H^1(X)} \leq C \|Tu\|_{L_r^2}, \quad u \in D(P_D). \quad (\text{DSP4})$$

The original P_D is retained; the symbol T always records the shift explicitly.

Hilbert representation, with both functional conventions. We first give the representation argument used by the form and kernel constructions. Let H be a complex Hilbert space with inner product linear in its first argument and let $\ell : H \rightarrow \mathbb{C}$ be bounded and linear. If $\ell = 0$, its representing vector is zero. Otherwise the closed affine set $V = \{v : \ell(v) = 1\}$ is nonempty. Its distance d from zero satisfies $d \geq \|\ell\|^{-1} > 0$. Choose $v_j \in V$ with $\|v_j\| \rightarrow d$. The midpoint belongs to V , so the parallelogram identity gives

$$\|v_j - v_k\|^2 = 2\|v_j\|^2 + 2\|v_k\|^2 - 4\|(v_j + v_k)/2\|^2 \leq 2\|v_j\|^2 + 2\|v_k\|^2 - 4d^2 \rightarrow 0.$$

Completeness and closedness give a limit $v \in V$ with $\|v\| = d$. For $z \in \ker \ell$, the inequality $\|v + tz\|^2 \geq d^2$, first for all real t and then for all purely imaginary t , forces $(v, z) = 0$. Since $u - \ell(u)v \in \ker \ell$, it follows that $(u, v) = \ell(u)d^2$. Thus $h = v/d^2$ satisfies

$$\ell(u) = (u, h), \quad \|\ell\| = \|h\|.$$

The norm equality follows from Cauchy–Schwarz and evaluation at $u = h/\|h\|$; uniqueness follows by testing the difference of two representing vectors against itself. For a bounded conjugate-linear functional F , apply this result to $\ell(u) = \overline{F(u)}$. Its vector w satisfies $F(u) = (w, u)$ with the same norm. This proves the exact forms of representation used below, including the equivalent inner product q_μ , without a spectral or orthonormal-basis theorem.

Proof of the realization. Integration by parts on compact interior tests gives the form

$$q(u, v) = \int_X r g^{jk} \partial_k u \overline{\partial_j v} dx + \int_X (g^{jk} \partial_j r + r b^k) \partial_k u \bar{v} dx + \int_X r c u \bar{v} dx. \quad (\text{DSP5})$$

Every term is continuous on $H_0^1 \times H_0^1$. Density extends symmetry from tests to that space. Positivity of the first term and boundedness of the other coefficients imply

$$q(u, u) \geq r_{\min} \theta \|\nabla u\|_2^2 - C \|u\|_2 \|\nabla u\|_2 - C \|u\|_2^2.$$

Young's inequality, $2ab \leq \varepsilon a^2 + \varepsilon^{-1} b^2$, gives a sufficiently large $\mu \geq 1$ for which $q_\mu = q + \mu(\cdot, \cdot)_r$ controls $\|u\|_{H^1}^2$ and also $\|u\|_{L_r^2}^2$. It is an inner product with an equivalent complete norm. Applying the Hilbert representation theorem in this norm to $v \mapsto (f, v)_r$ produces a unique $u \in H_0^1$ with $q_\mu(u, v) = (f, v)_r$ for every $v \in H_0^1$. Interior testing says $(P + \mu)u = f$ distributionally. Conversely, if $u \in H_0^1$ and $(P + \mu)u = f \in L_r^2$, the distributional identity extends to this form identity by density. Thus the variational operator is exactly T on (DSP3), and T maps its domain onto L_r^2 .

This domain is dense, since it contains $C_c^\infty(X)$. The form identity makes T symmetric. If $w \in D(T^*)$, solve $Tv = T^*w$. For every $u \in D(T)$, symmetry and adjointness give $(Tu, w - v)_r = 0$. Surjectivity of T forces $w = v \in D(T)$. Hence $T = T^*$, in particular it is closed. Coercivity gives the first estimate in (DSP4), and Cauchy–Schwarz gives

$$a \|u\|_{H^1}^2 \leq \|Tu\|_{L_r^2} \|u\|_{L_r^2} \leq C_r \|Tu\|_{L_r^2} \|u\|_{H^1},$$

which proves the second, including the case $u = 0$. The same form lower bound proves $T \geq I$. ■

2. Inhomogeneous Sobolev convention

For an integer $k \geq 0$, write

$$\overline{H}^k(X) = H^k(X), \quad \|u\|_{\overline{H}^k(X)}^2 = \sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^2(X)}^2. \quad (\text{DSP6})$$

The derivatives are weak derivatives. The bar indicates this inhomogeneous norm; it does not indicate closure of $C_c^\infty(X)$. In particular $\overline{H}^1(X)$ and $H_0^1(X)$ have different boundary conditions. Weighted and unweighted norms obey the exact comparison

$$r_{\min} \|u\|_2^2 \leq \|u\|_{L_r^2}^2 \leq r_{\max} \|u\|_2^2. \quad (\text{DSP7})$$

On a compact coordinate patch the same comparison uses the minimum and maximum there and the chart Jacobian. All constants involving these comparisons retain that dependence.

For reference, all local and transformed Sobolev symbols used below have the following fixed meaning. If $\Phi : U \rightarrow V$ is a smooth chart and $J_\Phi = |\det D\Phi^{-1}|$, then

$$\|u\|_{H_\Phi^s(U)}^2 := \sum_{|\alpha| \leq s} \int_V |D_y^\alpha (u \circ \Phi^{-1})(y)|^2 dy \quad (s \in \mathbb{N}),$$

while the weighted transformed norm is

$$\|u\|_{L^2_{r,\Phi}(U)}^2 = \int_V |u \circ \Phi^{-1}(y)|^2 (r \circ \Phi^{-1})(y) J_\Phi(y) dy.$$

For a compact $K \subset \overline{X}$, $H^s(K)$ denotes the fixed localized quantity obtained from a finite family of cutoffs $\zeta_a \in C_c^\infty(U_a)$ whose regions of value one cover K . The charts, cutoffs, and their larger neighborhoods remain part of this notation in each estimate. Coordinate changes of those same localized functions have the two-sided bounds proved below; arbitrary cutoffs with different outer supports need not give equivalent seminorms on all functions. For real noninteger $s \geq 0$, the localized coordinate spaces use the restriction norms (RT1)–(RT2). A finite partition covering the whole domain gives a norm equivalent to that restriction norm by the explicit extension and reconstruction below. On a smooth boundary chart, γu denotes the one-sided trace on $t = 0$, taking H^s continuously to $H^{s-1/2}$ for $s > 1/2$; H_0^1 is the kernel of this trace (equivalently the H^1 -closure of $C_c^\infty(X)$). For complete precision, put $\Omega_a = \Phi_a(U_a \cap X)$ and $v_a = ((\zeta_a u) \circ \Phi_a^{-1})|_{\Omega_a}$. The aggregate just specified is the sum

$$\|u\|_{H^s(K;\mathcal{A},\zeta)} = \sum_a N_{s,\Omega_a}(v_a), \quad N_{k,\Omega}(v) = \left(\sum_{|\alpha| \leq k} \int_\Omega |D_y^\alpha v|^2 dy \right)^{1/2} \quad (k \in \mathbb{N}_0), \quad (\text{DSP7a})$$

where at a noninteger $s \geq 0$ the symbol $N_{s,\Omega}$ instead means the restriction infimum in (RT2), with the whole-space norm (RT1). The abbreviation $\|u\|_{H^s(K)}$ retains the fixed atlas, cutoffs and domains Ω_a . It is a seminorm of functions on those fixed neighborhoods; it is not determined by their values on K alone. The integer formula keeps every coefficient in (DSP6) equal to one.

These definitions apply to every occurrence of transformed H^s , trace spaces, and local $H^s(K)$ below.

Real-order trace and boundary density

The following argument proves the real-order trace assertion, its fixed right inverse, the coordinate comparisons, and the zero-trace energy-space identity. Its numbered parts and RT formula labels belong to this argument. The derivative-sum norm (DSP6) retains its original coefficients.

Trace proof 1. Spaces and precise statement

Put $D_j = -i\partial_j$,

$$\hat{v}(\xi) = \int_{\mathbb{R}^N} e^{-ix \cdot \xi} v(x) dx, \quad v(x) = (2\pi)^{-N} \int e^{ix \cdot \xi} \hat{v}(\xi) d\xi,$$

and define the Bessel-potential norm by

$$\|v\|_{H^s(\mathbb{R}^N)}^2 = (2\pi)^{-N} \int_{\mathbb{R}^N} (1 + |\xi|^2)^s |\hat{v}(\xi)|^2 d\xi. \quad (\text{RT1})$$

For $s \geq 0$ and an open set X , $H^s(X)$ is the restriction space with norm

$$\|u\|_{H^s(X),\text{res}} = \inf\{\|U\|_{H^s(\mathbb{R}^n)} : U|_X = u\}. \quad (\text{RT2})$$

The kernel of restriction is closed: convergence in (RT1) implies convergence in L^2 , and a limit of functions zero almost everywhere on X is zero there. Thus (RT2) is a Hilbert quotient norm. Restriction is equality almost everywhere, or equivalently equality of distributions on X .

For integer $k \geq 0$, the actual squared norm in (DSP6) remains $\sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^2(X)}^2$. Already on the whole space its relation to (RT1) is the explicit identity

$$\|U\|_{H^k(\mathbb{R}^n)}^2 = \sum_{j=0}^k \binom{k}{j} \sum_{|\alpha|=j} \frac{j!}{\alpha!} \|D^\alpha U\|_2^2. \quad (\text{RT3})$$

On a smooth domain the restriction and weak-derivative spaces agree, with equivalent norms, by the integer reflection extension in Section 2 of Boundary energy, local inverses, and harmonic data and finite smooth localization. The coordinate estimates proved below justify that localization. Equation (RT3) specifies the coefficients of the whole-space comparison; no equality between the quotient norm and the derivative-sum norm on X is asserted.

Let X be bounded with smooth boundary. For every real $s > 1/2$ there is a unique bounded trace

$$\gamma_X : H^s(X) \longrightarrow H^{s-1/2}(\partial X) \quad (\text{RT4})$$

that agrees with boundary restriction for functions smooth on a neighborhood of $\bar{X}$. The surface space and its chart norm are defined in Trace proof 5, with an explicit comparison to surface measure. There is a bounded linear right inverse L_X , and one fixed choice of L_X works simultaneously at all these orders, with an order-dependent norm. In every localized flattened chart the trace is the limit of the positive slices in $H^{s-1/2}$, independent of the extension in (RT2). Finally,

$$\ker(\gamma_X : H^1(X) \rightarrow H^{1/2}(\partial X)) = \overline{C_c^\infty(X)}^{H^1(X)} = H_0^1(X). \quad (\text{RT5})$$

We now prove all assertions in (RT4)–(RT5).

Trace proof 2. Whole-space slices and quotient independence

Write $n = d + 1$, $x = (z, t)$, $\xi = (\zeta, \tau)$, and retain the inhomogeneous factor $\lambda(\zeta) = (1 + |\zeta|^2)^{1/2}$. For $s > 1/2$, put

$$C_s = \int_{\mathbb{R}} (1 + r^2)^{-s} dr = \frac{\sqrt{\pi} \Gamma(s - 1/2)}{\Gamma(s)}. \quad (\text{RT6})$$

For completeness, the gamma evaluation follows by inserting $(1 + r^2)^{-s} = \Gamma(s)^{-1} \int_0^\infty a^{s-1} e^{-a(1+r^2)} da$, applying Tonelli, and using $\int e^{-ar^2} dr = \sqrt{\pi/a}$. In particular

$$\int_{\mathbb{R}} (\lambda^2 + \tau^2)^{-s} d\tau = C_s \lambda^{1-2s}. \quad (\text{RT7})$$

For $U \in H^s(\mathbb{R}^{d+1})$, weighted Cauchy–Schwarz and Fubini define, for almost every ζ and every t ,

$$\widehat{T_t U}(\zeta) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{it\tau} \widehat{U}(\zeta, \tau) d\tau. \quad (\text{RT8})$$

The integral is absolutely convergent for those ζ . Equations (RT7)–(RT8) give

$$\begin{aligned} \|T_t U\|_{H^{s-1/2}(\mathbb{R}^d)}^2 &\leq (2\pi)^{-d-2} \int \lambda^{2s-1} C_s \lambda^{1-2s} \int (\lambda^2 + \tau^2)^s |\widehat{U}(\zeta, \tau)|^2 d\tau d\zeta \\ &= \frac{C_s}{2\pi} \|U\|_{H^s(\mathbb{R}^{d+1})}^2. \end{aligned} \quad (\text{RT9})$$

This is uniform for all real t .

Let $\mathcal{T}_h U(z, t) = U(z, t + h)$. Its Fourier multiplier is $e^{ih\tau}$, so dominated convergence in (RT1) proves $\mathcal{T}_h U \rightarrow U$ in H^s . Formula (RT8) gives $T_{t+h} U = T_t \mathcal{T}_h U$. Consequently (RT9) proves that $t \mapsto T_t U$ is continuous into $H^{s-1/2}(\mathbb{R}^d)$.

This continuous family is a representative of the actual slices of U . Indeed choose Schwartz $U_j \rightarrow U$ in H^s . Such density follows by approximating $\widehat{U}$ in the weighted L^2 space by smooth compactly supported functions; the inverse Fourier transforms are Schwartz. On any bounded interval I , (RT9) gives convergence of $T_t U_j$ to $T_t U$ in $L^2(I; L^2(\mathbb{R}^d))$, while $U_j \rightarrow U$ in $L^2(\mathbb{R}^{d+1})$. Fourier inversion identifies the slices for each Schwartz approximant, and uniqueness of an L^2 limit identifies them almost everywhere for U .

Suppose now that $U = 0$ almost everywhere on $\{t > 0\}$. The preceding identification implies $T_t U = 0$ for almost every $t > 0$; continuity implies this for every $t > 0$, and then for $t = 0$. Thus two full-space extensions of the same $u \in H^s(\mathbb{R}_+^{d+1})$ have the same T_0 , and even the same continuous slice family for $t \geq 0$. Define $\gamma_+ u = T_0 U$ using either extension and take the infimum in (RT9):

$$\sup_{t \geq 0} \|u(\cdot, t)\|_{H^{s-1/2}} \leq \left(\frac{C_s}{2\pi}\right)^{1/2} \|u\|_{H^s(\mathbb{R}_+^{d+1}), \text{res}}, \quad u(\cdot, t) \rightarrow \gamma_+ u \text{ in } H^{s-1/2} \quad (t \downarrow 0). \quad (\text{RT10})$$

The slice notation in (RT10) means this continuous representative. This argument proves quotient independence without presupposing a trace on the restriction space.

For $d = 0$, interpret $\mathbb{R}^0$ as one point with measure one, $\lambda = 1$, and its H^q norm as absolute value for every q . Equations (RT6)–(RT10) then are the same one-variable proof, with no tangential integration and with $(2\pi)^0 = 1$.

Trace proof 3. The same Poisson lift at every real order

Choose once and for all $\chi \in C^\infty(\mathbb{R})$ with $\chi(r) = 0$ for $r \leq -2$, $\chi(r) = 1$ for $r \geq -1$, and $0 \leq \chi \leq 1$, and put $\vartheta(r) = \chi(r)e^{-r}$. Then ϑ is Schwartz, $\vartheta(0) = 1$, and $\vartheta(r) = e^{-r}$ on $[0, \infty)$. For $f \in H^{s-1/2}(\mathbb{R}^d)$, define Ef by its full Fourier transform

$$\widehat{Ef}(\zeta, \tau) = \lambda(\zeta)^{-1} \widehat{\vartheta}(\tau/\lambda(\zeta)) \widehat{f}(\zeta). \quad (\text{RT11})$$

Change variables $\tau = \lambda\rho$ in (RT1). There is the exact identity

$$\|Ef\|_{H^s(\mathbb{R}^{d+1})}^2 = K_{s, \vartheta} \|f\|_{H^{s-1/2}(\mathbb{R}^d)}^2, \quad K_{s, \vartheta} = \frac{1}{2\pi} \int_{\mathbb{R}} (1 + \rho^2)^s |\widehat{\vartheta}(\rho)|^2 d\rho < \infty. \quad (\text{RT12})$$

Normal Fourier inversion, first for smooth data and then by (RT12), gives

$$\mathcal{F}_z(Ef)(\zeta, t) = \vartheta(\lambda t) \widehat{f}(\zeta), \quad \mathcal{F}_z(Lf)(\zeta, t) = e^{-\lambda t} \widehat{f}(\zeta) \quad (t > 0), \quad Lf = (Ef)|_{t>0}. \quad (\text{RT13})$$

In (RT8), $(2\pi)^{-1} \int \widehat{\vartheta}(\rho) d\rho = \vartheta(0) = 1$; hence $\gamma_+ Lf = f$. Equivalently, dominated convergence applied to $|e^{-\lambda t} - 1|^2 \lambda^{2s-1} |\widehat{f}|^2$ proves the same boundary limit. Equations (RT12)–(RT13) prove

$$\|Lf\|_{H^s(\mathbb{R}_+^{d+1}), \text{res}} \leq K_{s, \vartheta}^{1/2} \|f\|_{H^{s-1/2}}, \quad \gamma_+ L = I \quad \text{for every real } s > 1/2. \quad (\text{RT14})$$

The formula for L is exactly the positive-half-space lift in Section 2 of Boundary energy, local inverses, and harmonic data, formula (B4), independent of s . Its full-space extension (RT11) was chosen smooth across $t = 0$; none of the estimates assumes that an even reflection of the Poisson exponential has arbitrarily high regularity.

Trace proof 4. Fractional coordinate estimates with their dependencies

We prove the coordinate facts needed for both the domain and the surface. This part is in dimension $N \geq 1$. For $0 < \sigma < 1$, set

$$[v]_{\sigma,N}^2 = \iint_{\mathbb{R}^N \times \mathbb{R}^N} \frac{|v(x) - v(y)|^2}{|x - y|^{N+2\sigma}} dx dy.$$

Plancherel for translations and Tonelli give the exact identity

$$[v]_{\sigma,N}^2 = c_{N,\sigma} (2\pi)^{-N} \int |\xi|^{2\sigma} |\hat{v}(\xi)|^2 d\xi, \quad c_{N,\sigma} = \frac{2\pi^{N/2} \Gamma(1-\sigma)}{\sigma 4^\sigma \Gamma(N/2 + \sigma)}. \quad (\text{RT15})$$

Here is the constant calculation. The inner translation integral is $\int |e^{ih \cdot \xi} - 1|^2 |h|^{-N-2\sigma} dh$. Rotations and scaling reduce it to $|\xi|^{2\sigma}$ times its value at the first coordinate unit vector. Insert

$$|h|^{-N-2\sigma} = \Gamma(N/2 + \sigma)^{-1} \int_0^\infty a^{N/2+\sigma-1} e^{-a|h|^2} da.$$

The Gaussian integral of $2 - 2\cos h_1$ is $2(\pi/a)^{N/2}(1 - e^{-1/(4a)})$. Substituting $r = 1/(4a)$ leaves $2\pi^{N/2} 4^{-\sigma} \Gamma(N/2 + \sigma)^{-1} \int_0^\infty r^{-\sigma-1} (1 - e^{-r}) dr$. Integration by parts evaluates the final integral as $\Gamma(1-\sigma)/\sigma$; its boundary terms vanish because $0 < \sigma < 1$. This proves (RT15), including its constants. It holds for every L^2 function, with both sides allowed to be infinite, by Plancherel for the L^2 translation difference and Tonelli.

Let $m \geq 0$ be an integer and define

$$\mathcal{E}_{m,\sigma}(v) = \sum_{|\alpha| \leq m} \|D^\alpha v\|_2^2 + c_{N,\sigma}^{-1} \sum_{|\alpha|=m} [D^\alpha v]_{\sigma,N}^2.$$

Fourier transformation, the multinomial formula, and (RT15) give

$$\frac{1}{2^{m+\sigma} m!} \|v\|_{H^{m+\sigma}}^2 \leq \mathcal{E}_{m,\sigma}(v) \leq (m+2) \|v\|_{H^{m+\sigma}}^2. \quad (\text{RT16})$$

Indeed the multiplier of $\mathcal{E}$ is $P_m(\xi) + |\xi|^{2\sigma} Q_m(\xi)$, where $P_m = \sum_{|\alpha| \leq m} \xi^{2\alpha}$ and $Q_m = \sum_{|\alpha|=m} \xi^{2\alpha}$. The inequalities $|\xi|^{2m}/m! \leq Q_m \leq |\xi|^{2m}$ and $P_m \leq (m+1) \max(1, |\xi|^{2m})$, split at $|\xi| = 1$, prove exactly (RT16). The case $m = 0$ uses $0! = 1$.

We next give a local composition bound that does not assume the desired fractional invariance. Let $\Psi : V \rightarrow W$ be a smooth diffeomorphism, and let $b \in C_c^\infty(V_0)$, where $V_0 \Subset V$. Choose the smaller coordinate set so that for $y, z \in V_0$

$$L^{-1}|y - z| \leq |\Psi(y) - \Psi(z)| \leq L|y - z|, \quad J = \sup_{\Psi(V_0)} |\det D\Psi^{-1}| < \infty, \quad (\text{RT17})$$

with $L \geq 1$. Smooth coordinate maps admit such sets: on a sufficiently small ball about any point, the derivative differs from its invertible value at the center by less than half that value's smallest singular value. Integrating the derivative on line segments proves (RT17); finitely many such smaller sets cover a prescribed compact support. Thus using several such sets entails only a finite smooth partition.

Write $B = \|b\|_\infty$, $M = \|\nabla b\|_\infty$, $\delta = \text{dist}(\text{supp } b, \mathbb{R}^N \setminus V_0) > 0$, and let ω_{N-1} be the area of the unit sphere in $\mathbb{R}^N$. Set

$$A_{b,\sigma} = \omega_{N-1} \left(\frac{M^2}{2-2\sigma} + \frac{4B^2}{2\sigma} \right).$$

For $v(y) = b(y)f(\Psi(y))$ on V_0 , extended by zero outside, change of variables gives

$$\|v\|_2^2 \leq B^2 J \|f\|_2^2,$$

and direct separation of differences gives

$$[v]_{\sigma,N}^2 \leq 2B^2 L^{N+2\sigma} J^2 [f]_{\sigma,N}^2 + \left(2A_{b,\sigma} J + \frac{\omega_{N-1}}{\sigma} \delta^{-2\sigma} B^2 J\right) \|f\|_2^2. \quad (\text{RT18})$$

To verify every term, on $V_0 \times V_0$ use

$$|b(y)f(\Psi(y)) - b(z)f(\Psi(z))|^2 \leq 2B^2 |f(\Psi(y)) - f(\Psi(z))|^2 + 2|b(y) - b(z)|^2 |f(\Psi(z))|^2.$$

Two Jacobians and (RT17) bound the first integral. For the second, extend b smoothly by zero; then $|b(y) - b(z)| \leq \min(M|y - z|, 2B)$. Integration on $|y - z| \leq 1$ and $|y - z| > 1$ yields $A_{b,\sigma}$, followed by one Jacobian. Finally the two cross regions with one point outside V_0 contribute at most $2\|v\|_2^2 \omega_{N-1} \int_\delta^\infty r^{-1-2\sigma} dr$. This is the last term of (RT18). Thus no boundary term in the zero extension of a localized composition was omitted.

For a cutoff $a \in C_c^\infty(V_0)$, apply the usual chain and product rules to $a(f \circ \Psi)$. In ordinary derivative notation their exact coefficients can be generated recursively by

$$b_{0,0} = a, \quad b_{\alpha+e_j,\beta} = \partial_j b_{\alpha,\beta} + \sum_{\ell=1}^N b_{\alpha,\beta-e_\ell} \partial_j \Psi_\ell, \quad \partial^\alpha (a(f \circ \Psi)) = \sum_{|\beta| \leq |\alpha|} b_{\alpha,\beta} ((\partial^\beta f) \circ \Psi). \quad (\text{RT19})$$

Absent coefficients are zero; for each multiindex fix any ordered sequence of the indicated differentiations. The product rule proves this recursion by induction. The norms of $D^\alpha v$ and $\partial^\alpha v$ are equal, so these ordinary derivative identities do not change the convention $D = -i\partial$.

For integer order, change of variables applied to each term of (RT19) gives the bounded composition operator in H^m . For order $m + \sigma$, use (RT18) on each term of each order- m derivative, then (RT16). For every $|\beta| \leq m$, Plancherel also gives

$$\|D^\beta f\|_2^2 \leq \|f\|_{H^{m+\sigma}}^2, \quad [D^\beta f]_{\sigma,N}^2 \leq c_{N,\sigma} \|f\|_{H^{m+\sigma}}^2.$$

Using $|\sum_{j=1}^r v_j|^2 \leq r \sum |v_j|^2$ for each finite sum completes the estimate

$$\|a(f \circ \Psi)\|_{H^r(\mathbb{R}^N)} \leq C_{r,a,\Psi,V_0} \|f\|_{H^r(\mathbb{R}^N)}, \quad r \geq 0, \quad (\text{RT20})$$

where the left side is the zero extension just used. Its squared constant is obtained by the finite sums in (RT19), (RT18), and (RT16). In particular it depends only on N, m, σ , the displayed L, J, δ , the derivatives of a through order $m + 1$, and those of Ψ through order $m + 1$. At an integer order no fractional constant occurs and derivatives through order m suffice. Applying the same proof to inverse coordinate maps gives the reverse local comparisons. General compactly supported coordinate changes follow by the finite subdivision explained after (RT17).

All these identities were initially proved on smooth inputs. Density of Schwartz functions in (RT1), already proved in Trace proof 2, extends (RT20) to H^r . The resulting extension is the actual composition almost everywhere, since it converges in L^2 on the support by the Jacobian bound. Thus its weak derivatives are also those supplied by (RT19).

Multiplication by a fixed compact smooth cutoff is (RT20) with $\Psi = I$. Restrictions of $C_c^\infty(\mathbb{R}^N)$ are dense in every quotient (RT2): after Schwartz approximation, multiply a fixed Schwartz function by cutoffs $\theta(x/R)$. The difference tends to zero in every integer H^k by the product rule, the rapid decay of all derivatives, and dominated convergence; choosing $k \geq r$ and using (RT1) gives convergence in H^r . Approximate a chosen full-space extension and restrict. This also proves density of functions smooth on a neighborhood of $\bar{X}$ in $H^r(X)$.

Trace proof 5. Surface norms and atlas independence

For $n \geq 2$, put $d = n - 1$. Fix a finite smooth boundary atlas $\kappa_a : O_a \subset \mathbb{R}^d \rightarrow \partial X$, and a nonnegative smooth partition λ_a on ∂X subordinate to relatively compact parts of these charts, with $\sum_{a=1}^A \lambda_a = 1$. For $q \geq 0$, define

$$\|f\|_{H^q(\partial X), \mathcal{A}}^2 = \sum_{a=1}^A \|((\lambda_a f) \circ \kappa_a)^0\|_{H^q(\mathbb{R}^d)}^2, \quad (\text{RT21})$$

where superscript zero means extension by zero outside O_a . The surface space consists of $L^2(dS)$ functions with finite (RT21).

Let $J_a(z) = \sqrt{\det(D\kappa_a(z)^T D\kappa_a(z))}$. This is positive and smooth. If $j_{\min}, j_{\max}$ are the minimum and maximum of all J_a on the compact supports concerned, change of variables gives the exact order-zero formula and bounds

$$\begin{aligned} \|f\|_{H^0(\partial X), \mathcal{A}}^2 &= \int_{\partial X} |f(x)|^2 \sum_{a: x \in \kappa_a(O_a)} \frac{\lambda_a(x)^2}{J_a(\kappa_a^{-1}(x))} dS(x), \\ \frac{1}{Aj_{\max}} \|f\|_{L^2(dS)}^2 &\leq \|f\|_{H^0(\partial X), \mathcal{A}}^2 \leq \frac{1}{j_{\min}} \|f\|_{L^2(dS)}^2. \end{aligned} \quad (\text{RT22})$$

Terms with $\lambda_a = 0$ are zero. The inequalities use $A^{-1} \leq \sum \lambda_a^2 \leq 1$. If a positive surface weight w is used, its exact measure comparison is $w_{\min} \|f\|_{L^2(dS)}^2 \leq \|f\|_{L^2(wdS)}^2 \leq w_{\max} \|f\|_{L^2(dS)}^2$; substituting these bounds in (RT22) retains both the weight and the chart Jacobians.

To prove atlas independence, take another chart partition $(\tilde{\kappa}_b, \tilde{\lambda}_b)$. In a b -chart write

$$(\tilde{\lambda}_b f) \circ \tilde{\kappa}_b = \sum_a (\tilde{\lambda}_b \circ \tilde{\kappa}_b) [((\lambda_a f) \circ \kappa_a) \circ (\kappa_a^{-1} \circ \tilde{\kappa}_b)]. \quad (\text{RT23})$$

For each overlap insert an extra smooth cutoff equal to one on the compact intersection of the two original supports and supported inside the overlap. The summands are then exactly the localized compositions of (RT20), in dimension d , at order q . The number of summands is finite. Equation (RT20) therefore bounds the new norm by the old; exchanging the atlases proves the reverse bound. The constants have exactly the dependence described in Trace proof 4, including the overlap cutoffs and their distances to chart edges.

For completeness, (RT21) defines a complete space: a Cauchy sequence converges in $L^2(dS)$ by (RT22); each localized coordinate sequence converges in $H^q(\mathbb{R}^d)$, whose L^2 limit is the localization of that same surface function by change of variables. The limits therefore have finite (RT21), and convergence holds in that norm. Smooth surface functions are dense. To see this, each compactly supported coordinate function $((\lambda_a f) \circ \kappa_a)^0$ can be convolved with a smooth approximate identity and then multiplied by a fixed cutoff equal to one near its original support. Approximation in H^q follows from Fourier dominated convergence for convolution and bounded cutoff multiplication (RT20); for sufficiently small convolution radius the support remains in O_a . Push these smooth compact functions to the surface and sum over a . Formula (RT23) and (RT20) show convergence in the surface norm, and the limiting sum is $\sum_a \lambda_a f = f$.

In dimension $n = 1$, a compact smooth boundary is a finite zero-dimensional manifold: its single-point coordinate neighborhoods form an open cover, so compactness gives finitely many points. Define every $H^q(\partial X)$ as functions on these points with the sum of squared absolute values. Surface measure is counting measure. There are no fractional coordinate changes to prove in dimension zero.

Trace proof 6. Curved-boundary trace, independence, and lifts

Take the fixed boundary atlas from Trace proof 5 as the restriction of ambient smooth flattenings $\Phi_a : U_a \rightarrow V_a \subset \mathbb{R}^{d+1}$, with $\Phi_a(X \cap U_a) = V_a \cap \{t > 0\}$ and $\kappa_a(z) = \Phi_a^{-1}(z, 0)$. Extend each λ_a to

a compactly supported smooth ambient cutoff $\eta_a \in C_c^\infty(U_a)$ with $\eta_a|_{\partial X} = \lambda_a$. This is achieved by multiplying its coordinate function on $t = 0$ by a smooth normal cutoff equal to one near zero. All supports remain in the chosen coordinate patches.

For a full-space H^s extension U of u , set

$$v_a(y) = \begin{cases} (\eta_a U)(\Phi_a^{-1}(y)), & y \in V_a \text{ (the chart range)}, \\ 0, & \text{elsewhere.} \end{cases} \quad (\text{RT24})$$

By (RT20), $\|v_a\|_{H^s} \leq C_a \|U\|_{H^s}$. Its flat trace $T_0 v_a$ is bounded by (RT9). For smooth U , it is precisely $(\lambda_a U|_{\partial X}) \circ \kappa_a$, extended by zero.

Approximate a general U in H^s by C_c^∞ functions. The traces of these approximants form a Cauchy sequence in the surface norm (RT21), because

$$\|U|_{\partial X}\|_{H^{s-1/2}(\partial X), \mathcal{A}}^2 \leq \frac{C_s}{2\pi} \sum_a C_a^2 \|U\|_{H^s(\mathbb{R}^n)}^2 \quad (\text{RT25})$$

holds for their differences. Completeness gives a limit. Bounded localization and (RT20) show that its a -coordinate component is exactly $T_0 v_a$, so it is independent of the approximating sequence.

If U vanishes on X , each localized function v_a in (RT24) vanishes on the entire positive half-space: inside its chart this follows from the flattening, and outside the chart it follows from its zero definition. Trace proof 2 gives $T_0 v_a = 0$. Hence the resulting surface limit is zero by (RT21). It follows that the boundary value depends only on $u = U|_X$. Taking the infimum over U in (RT25) proves (RT4) with

$$\|\gamma_X u\|_{H^{s-1/2}(\partial X), \mathcal{A}} \leq \left(\frac{C_s}{2\pi} \sum_a C_a^2 \right)^{1/2} \|u\|_{H^s(X), \text{res}}. \quad (\text{RT26})$$

Smooth restrictions are dense by Trace proof 4, so this continuous extension of smooth restriction is unique. Consequently changing the flattenings or partition yields exactly the same trace, with the equivalent surface norms of Trace proof 5. The positive-slice limit in each cutoff chart is exactly (RT10) applied to $v_a|_{t>0}$. This proves the asserted local one-sided interpretation at every real $s > 1/2$.

Here are also the precise local Sobolev comparisons implicit in the notation of Section 2. Choose a finite partition of unity on a neighborhood of $\overline{X}$, subordinate to boundary and interior patches, and in each patch take the corresponding localized coordinate restriction norm of order s . Such an ambient partition can retain boundary values λ_a : choose the nonnegative extensions above, divide by their positive sum near the boundary, multiply the resulting functions by a collar cutoff equal to one on a smaller boundary neighborhood, and partition the remaining interior term. Equation (RT20), applied to any extension U , bounds the finite localized norm by $C\|u\|_{H^s(X), \text{res}}$ after taking the infimum. Conversely extend each localized coordinate restriction to the whole coordinate space with norm within ε of its quotient infimum. Multiply by a fixed slightly larger cutoff equal to one on the original localized support and pull back by the inverse chart, extending by zero beyond that chart. Equation (RT20) bounds each resulting global H^s function. The sum restricts to u , since the original cutoffs sum to one. The triangle inequality and then $\varepsilon \downarrow 0$ bound (RT2) by the finite localized norm. This proves both directions, with cutoff, Jacobian, and chart constants retained. For an estimate near a compact subset, fix the localizing functions and larger patch first and apply this construction to those localized functions; the constants retain both choices. No equivalence between seminorms on unrelated larger neighborhoods is needed. Interior pieces are treated by whole-space coordinate charts and the same proof.

To construct a right inverse, first choose once and for all $\rho_a \in C_c^\infty(V_a)$ whose value on $t = 0$ is one on a neighborhood of the fixed compact set $\text{supp}((\lambda_a \circ \kappa_a)^0)$. These cutoffs depend only on the atlas and

partition, not on the datum or the Sobolev order. For $f \in H^{s-1/2}(\partial X)$, put $f_a = ((\lambda_a f) \circ \kappa_a)^0$, whose support lies in that fixed set, and take $E f_a$ from (RT11). Define the global function

$$W_a(x) = \begin{cases} (\rho_a E f_a)(\Phi_a(x)), & x \in U_a, \\ 0, & x \notin U_a, \end{cases} \quad L_X f = \sum_a W_a \Big|_X. \quad (\text{RT27})$$

Equations (RT12), (RT20), and the finite sum give

$$\|L_X f\|_{H^s(X), \text{res}} \leq K_{s, \vartheta}^{1/2} \left(\sum_a C'_{a,s}{}^2 \right)^{1/2} \|f\|_{H^{s-1/2}(\partial X), \mathcal{A}}, \quad (\text{RT28})$$

where $C'_{a,s}$ is the bounded composition-and-cutoff constant for the a -term. For smooth f , direct evaluation and $\vartheta(0) = 1$ give $\gamma_X W_a = \lambda_a f$; hence $\gamma_X L_X f = f$. Surface density and (RT26), (RT28) extend this identity to all f . The choices of ϑ , all cutoffs, and the atlas were fixed independently of s . Therefore the operator (RT27) is a single right inverse on all real scales $s > 1/2$, with the displayed order-dependent bounds. This construction also works at each boundary point when $n = 1$, using the one-variable instance of (RT11).

Trace proof 7. The zero-trace energy space

For clarity we prove (RT5) with the derivative-sum H^1 norm of (DSP6), which is equivalent to (RT2). The inclusion of the closure into the kernel follows immediately from the trace bound and the zero trace of every compact interior smooth function.

For the reverse inclusion, first work on the flat half-space with a compactly supported $v \in H^1(\mathbb{R}_+^{d+1})$ whose trace is zero. Write v^0 for zero extension. Smooth approximation by restrictions of whole-space tests, proved in Trace proof 4, and integration by parts give the distribution formulas

$$\partial_t v^0 = (\partial_t v)^0 + (\gamma_+ v) \otimes \delta_0, \quad \partial_{z_j} v^0 = (\partial_{z_j} v)^0. \quad (\text{RT29})$$

Indeed, for a smooth approximant and a compact smooth test ψ , integration on $t > 0$ yields $-\int v \partial_t \psi = \int (\partial_t v) \psi + \int v(z, 0) \psi(z, 0) dz$. The volume terms converge by H^1 convergence; the boundary term converges by (RT10) at $s = 1$, and the test boundary restriction is smooth compactly supported. This proves the normal formula, with its positive sign. Tangential integration has no boundary term and proves the other formulas. Since the trace is zero, every derivative on the right of (RT29) is L^2 , so $v^0 \in H^1(\mathbb{R}^{d+1})$.

Set $v_\varepsilon(z, t) = v^0(z, t - \varepsilon)$. It vanishes for $t < \varepsilon$, and Fourier dominated convergence gives $v_\varepsilon \rightarrow v^0$ in H^1 . Convolve it with a compact smooth mollifier of radius less than $\varepsilon/2$, choosing its radius sufficiently small that the H^1 convolution error is less than ε . The resulting smooth function is supported in $t \geq \varepsilon/2$ and converges to v^0 in H^1 . The original compact support and sufficiently small translations and radii keep it within any fixed slightly larger coordinate neighborhood. A fixed cutoff equal to one near the original support, if needed, preserves convergence by (RT20). Thus the restrictions approximate v by compact interior smooth functions.

For a zero-trace $u \in H^1(X)$, use a finite smooth partition subordinate to boundary and interior patches. Each boundary-localized pullback has zero flat trace by (RT24)–(RT26) and is approximated by the preceding construction. Pull those approximants back by the smooth flattening and extend by zero beyond its compact patch. They are $C_c^\infty(X)$, and (RT20) at order one proves their convergence in $H^1(X)$. The interior-localized pieces are approximated by ordinary mollification with radius smaller than their distance from the boundary. Summing the finite approximations proves that u belongs to the closure in (RT5).

Finally, (RT29) and this localization show the matching support statement when it is used: a zero-trace u has $u^0 \in H^1(\mathbb{R}^n)$, supported in $\bar{X}$. Conversely, if $U \in H^1(\mathbb{R}^n)$ is supported in $\bar{X}$, its localized pullbacks vanish on the negative side. The continuous slice family in Trace proof 2 has zero negative-side limit, so its common value at zero is zero. Thus $\gamma_X(U|_X) = 0$. Smooth boundaries have Lebesgue measure zero by their finite graph cover and Fubini, so zero extension introduces no unspecified values on a set of positive measure. This completes the proof of the full real-order trace theorem and the exact energy-kernel assertion.

3. Boundary regularity

For every integer $k \geq 0$,

$$u \in D(P_D), \quad Pu \in \bar{H}^k(X) \implies u \in \bar{H}^{k+2}(X), \quad \|u\|_{\bar{H}^{k+2}} \leq C_k(\|Pu\|_{\bar{H}^k} + \|u\|_{L^2_r}). \quad (\text{DSP8})$$

For the explicitly shifted operator this becomes

$$\|u\|_{\bar{H}^{k+2}} \leq C_{k,\mu} \|Tu\|_{\bar{H}^k}, \quad u \in D(T), \quad Tu \in \bar{H}^k. \quad (\text{DSP9})$$

Proof. Begin with the known $u \in H^1_0$. In each flattened boundary chart the transformed leading matrix is real positive definite and smooth, and all transformed lower coefficients are smooth and bounded on a smaller patch. The density changes by the positive Jacobian. The trace remains zero by the trace and coordinate-change construction in Section 2 of Boundary energy, local inverses, and harmonic data. Thus Section 6 of Boundary energy, local inverses, and harmonic data applies, including its bounded complex lower terms, and yields the local H^2 estimate in terms of $Pu \in L^2$ and $u \in H^1$. Its proof first tangentially mollifies an H^1 function, recovers its second normal derivative from the equation, and only then applies the strong estimate; it makes no assumption of the sought H^2 regularity. The interior theorem applies on interior patches. A finite cover gives

$$\|u\|_{H^2} \leq C(\|Pu\|_2 + \|u\|_{H^1}).$$

The form estimate (DSP4), with $Tu = Pu + \mu u$, bounds the final term by $C(\|Pu\|_2 + \|u\|_2)$. This proves $k = 0$ in (DSP8).

Here is the full higher-order induction, including the normal sign. In flat coordinates write the transformed expression, renaming its coefficients locally, as

$$P = \sum_{j,k} g^{jk} D_j D_k + \sum_k \ell^k D_k + c, \quad \ell^k = \sum_j D_j g^{jk} + i b^k \quad (\text{DSP10})$$

when it is written in the divergence form (DSP2). A coordinate change can modify the displayed ℓ^k , but preserves this positive sign of the principal $D_j D_k$ terms. The exact equation for the pure normal derivative is

$$D_n^2 u = (g^{nn})^{-1} \left(Pu - \sum_{(j,k) \neq (n,n)} g^{jk} D_j D_k u - \sum_k \ell^k D_k u - cu \right). \quad (\text{DSP11})$$

Assume inductively that $u \in H^{k+1}$ on a slightly larger chart and $Pu \in H^k$. For a purely tangential multiindex β of length k , $D_z^\beta u \in H^1$ has zero trace. For each monomial $a_\alpha D^\alpha$ of P , the product rule reads

$$[D_z^\beta, a_\alpha D^\alpha]u = \sum_{0 < \delta \leq \beta} \binom{\beta}{\delta} (D_z^\delta a_\alpha) D^{\alpha+\beta-\delta} u. \quad (\text{DSP12})$$

The derivatives on u have order at most $k+1$. To make the normal induction explicit, write a target multiindex as $\gamma = (\gamma', q)$, $|\gamma| = k+2$, with $q \geq 3$, and set $\delta = (\gamma', q-2)$, so $|\delta| = k$. Apply D^δ to (DSP11). For each coefficient a in the right-hand side, Leibniz gives

$$D^\delta(a D^\eta u) = a D^{\delta+\eta} u + \sum_{0 < \zeta \leq \delta} \binom{\delta}{\zeta} (D^\zeta a) D^{\delta-\zeta+\eta} u. \quad (\text{DSP12a})$$

In the first term, the Pu contribution is $D^\delta Pu$, bounded by $\|Pu\|_{H^k}$. For every principal term with $(j, \ell) \neq (n, n)$, the derivative $D^{\delta+e_j+e_\ell} u$ has at most $q-1$ normal factors (at most $q-2$ if both indices are tangential); the lower-order terms have at most $q-1$ as well. These are controlled by the induction hypothesis on q . Every coefficient-hit term in the sum has

$$|\delta - \zeta + \eta| \leq k+1$$

because $|\zeta| \geq 1$ and $|\eta| \leq 2$; hence it is controlled by the already established H^{k+1} bound. If $G_{k,q}$ denotes the L^2 -norm of all order- $k+2$ derivatives with at most q normals on a fixed patch, this gives

$$G_{k,q} \leq C_{k,q} (\|Pu\|_{H^k(U_1)} + \|u\|_{H^{k+1}(U_1)} + G_{k,q-1}), \quad (\text{DSP12b})$$

where $U_0 \Subset U_1$ are nested patches and $C_{k,q}$ depends only on the ellipticity lower bound and coefficient C^{k+1} -norms on U_1 . Starting with $q=2$ from the tangential estimate and iterating (DSP12b) through $q=k+2$ proves every order- $k+2$ derivative. In dimension one, (DSP11) is the same recurrence with no tangential base case.

For cutoffs χ , the exact commutator is

$$[P, \chi]u = -2g^{jk}(\partial_j \chi) \partial_k u + [-\partial_j(g^{jk} \partial_k \chi) + b^j \partial_j \chi]u.$$

Thus, for nested $U_0 \Subset U_1$ and every k ,

$$\|[P, \chi]u\|_{H^k(U_0)} \leq C_{k,\chi} \|u\|_{H^{k+1}(U_1)}, \quad C_{k,\chi} \leq C \left(\max_{|\rho| \leq k+1} \|D^\rho g\|_\infty, \max_{|\rho| \leq k} \|D^\rho(b, c)\|_\infty, \max_{|\rho| \leq k+2} \|D^\rho \chi\|_\infty \right). \quad (\text{DSP12c})$$

There are finitely many nested patches for each fixed k ; summing (DSP12b) and (DSP12c) over a finite partition of unity gives (DSP8). For example, in divergence coordinates the first commutator is exactly

$$[D_j, P]u = -\partial_\ell((D_j g^{\ell m}) \partial_m u) + (D_j b^m) \partial_m u + (D_j c)u. \quad (\text{DSP13})$$

To obtain (DSP9), apply the argument to the expression $P + \mu$. At $k=0$, (DSP4) removes the L^2 remainder. At higher orders the same inequality applies with $P + \mu$ in place of P , and $\|u\|_2 \leq C\|Tu\|_2$ removes it again. Thus no unshifted inverse is asserted when P_D has a kernel. ■

The exact first domain on a $C^{1,1}$ boundary

The first-domain conclusion has weaker boundary hypotheses than the smooth bootstrap. Keep the original expression (DSP2), its density r , formal symmetry, and coefficient bounds on a neighborhood of the compact closure. Allow the bounded domain X to have $C^{1,1}$ boundary. Then the realization (DSP3), coercive shift (DSP4), and compact positive inverse still exist, and

$$D(P_D) = H^2(X) \cap H_0^1(X), \quad \|u\|_{H^2(X)} \leq C(\|P_D u\|_{L_r^2} + \|u\|_{L_r^2}). \quad (\text{DSP15a})$$

Here H^2 keeps the weak derivative-sum norm (DSP6). This statement does not extend the all-order boundary bootstrap to this boundary class.

Proof. The form construction (DSP5) uses bounded coefficients, positive lower bounds and the definition of H_0^1 by compact interior approximation; it uses no second boundary derivative. It therefore proves (DSP3)–(DSP4) on this domain. On a compact boundary chart write, in the original rigid coordinates,

$$F(y', t) = (y', t + h(y')), \quad h \in C^{1,1}, \quad J_F = |\det DF| = 1, \quad v = u \circ F. \quad (\text{DSP15b})$$

Every coefficient contribution survives the weak change of variables. Put $M(y) = (DF(y))^{-1} = D(F^{-1})(F(y))$, and use ordinary derivatives in this formula. The exact pulled-back operator is

$$\begin{aligned} (Pu) \circ F &= -J_F^{-1} \partial_a \left[J_F (M(g \circ F) M^T)^{ab} \partial_b v \right] \\ &\quad + (b^j \circ F) M_{aj} \partial_a v + (c \circ F) v, \quad \tilde{r} = (r \circ F) J_F. \end{aligned} \quad (\text{DSP15c})$$

One can verify (DSP15c) first in the divergence weak pairing: change the volume variable, use $\nabla_x u \circ F = M^T \nabla_y v$, and then move the weak divergence back to the test. This proves the identity for H^1 functions without assuming their second derivatives. The transformed principal matrix is uniformly positive and Lipschitz on the compact patch; its positive bound follows from $\xi^T M(g \circ F) M^T \xi \geq \theta |M^T \xi|^2$. Its weak first derivatives contain every derivative of the matrix $(DF)^{-1}$ and of $g \circ F$, and are bounded. The first-order and zeroth-order coefficients in (DSP15c) are bounded, including the derivatives obtained on expanding its divergence. The positive transformed density retains its Jacobian.

The trace-zero condition needed here is also justified at this regularity. The H^1 chart pullback is bounded by the first chain rule and Jacobian bounds. The flat H^1 trace therefore gives a boundary trace of v . Compact interior approximants to $u \in H_0^1(X)$ have zero flat trace, so continuity gives $\gamma_+ v = 0$. The boundary coordinate comparison for $H^{1/2}$ uses only the Lipschitz distances, bounded Jacobians and bounded Lipschitz cutoffs in (RT18), with $m = 0, \sigma = 1/2$; none of its terms requires a smooth chart. Conversely a zero trace yields a zero extension in flat H^1 by (RT29). Translate it inward and mollify as in Trace proof 7. Pulling these compact interior approximants back gives H^1 functions supported a positive distance from ∂X ; ordinary interior mollification then gives $C_c^\infty(X)$ approximants. Thus the trace kernel is exactly $H_0^1(X)$, rather than an assumed boundary condition on an unknown representative.

The local H^1 -to- H^2 theorem in Section 6 of Boundary energy, local inverses, and harmonic data, together with its complete weak chart argument (BR1)–(BR3), applies to (DSP15c), with its bounded complex lower terms. It gives $v \in H^2$ locally up to the wall. Passing back uses the full weak second chain rule

$$\partial_i \partial_j (v \circ F^{-1}) = \sum_{a,b} (\partial_a \partial_b v) \circ F^{-1} \partial_i (F^{-1})_a \partial_j (F^{-1})_b + \sum_a (\partial_a v) \circ F^{-1} \partial_i \partial_j (F^{-1})_a. \quad (\text{DSP15d})$$

The last sum is bounded in L^2 , since the weak second derivatives of F^{-1} are bounded. Its validity for H^2 inputs follows by the flat reflection extension, smooth approximation and the weak Lipschitz product rule, as proved in (BR2). Interior estimates, a finite cover and (DSP4) now give the estimate in

(DSP15a). This proves the forward domain inclusion. Conversely $u \in H^2 \cap H_0^1$ has $Pu \in L_r^2$ by the original bounded coefficient expansion, so (DSP3) gives the reverse inclusion with exactly the same action.

The zero extension of an H_0^1 function is H^1 with the same derivative-sum norm, directly by the defining compact interior approximation. The compact-support Fourier proof in Section 5 thus applies without a smooth boundary assumption. Its compact positive inverse consequently gives the same eigenbasis construction (DSP16), with all weights and shifts retained. On a $C^{1,1}$ boundary these eigenfunctions have the H^2 regularity just proved; all-order boundary smoothness in the earlier statement continues to require its original smooth boundary. The graph $h(z) = z|z|$ in (BR7) gives a local example where this first-domain proof applies while the chart is not C^2 . Its exact matrix and weak drift are shown in the accompanying figure. ■

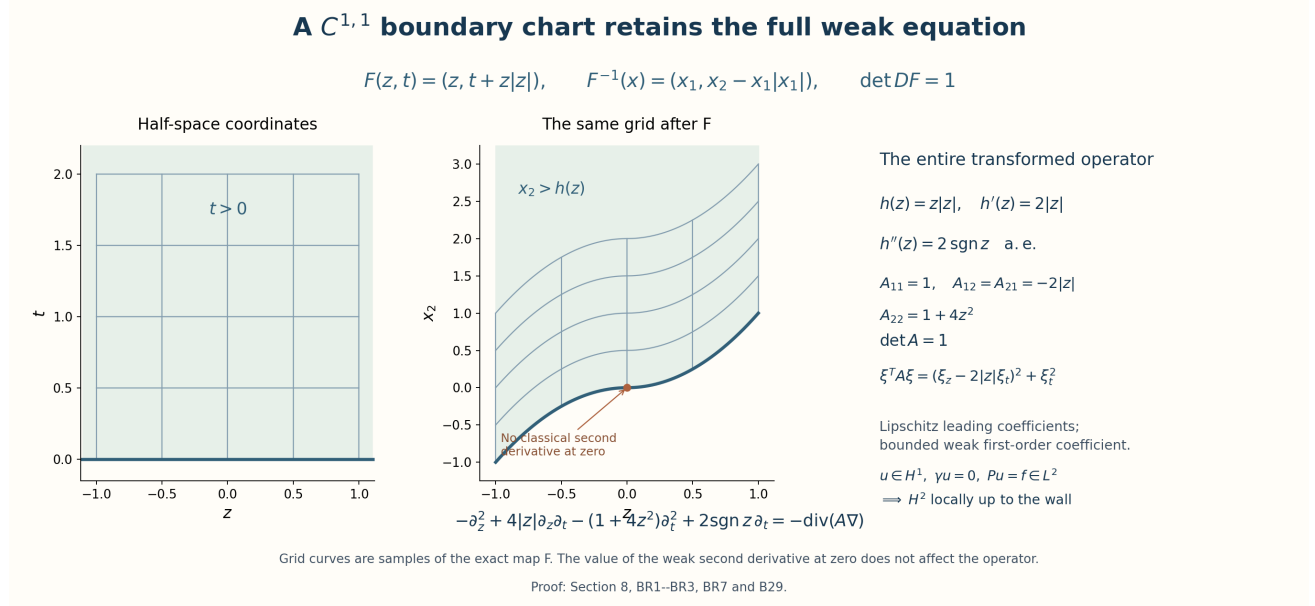


Figure 1: The original graph chart $F(z, t) = (z, t + z|z|)$, its full pulled-back matrix and its weak differential expression. The figure illustrates (DSP15b)–(DSP15d) through the exact local example (BR7) in the boundary lesson; it does not depict an all-order smooth boundary.

4. Domains of powers

For $m \geq 1$, powers use the operator domain, recursively:

$$D(T^m) = \{u \in D(T^{m-1}) : T^{m-1}u \in D(T)\}.$$

The scalar shift gives $D(T^m) = D(P_D^m)$. Indeed, induction and the binomial identities for $T = P_D + \mu I$ and $P_D = T - \mu I$ give both inclusions, retaining every lower power in its domain. Moreover,

$$\|u\|_{\overline{H}^{2m}} \leq C_m \|T^m u\|_{L_r^2}, \quad u \in D(T^m), \quad (\text{DSP14})$$

and the exact boundary-domain description is

$$D(P_D^m) = \{u \in H^{2m}(X) : \gamma(P^j u) = 0, \quad 0 \leq j < m\}. \quad (\text{DSP15})$$

Proof. The $m = 1$ estimate is (DSP9). If $u \in D(T^m)$, then $Tu \in D(T^{m-1})$, so induction gives $Tu \in H^{2m-2}$ with norm at most $C\|T^m u\|_2$; apply (DSP9) at $k = 2m - 2$. This proves (DSP14). Each $P_D^j u$, $j < m$, lies in $D(P_D)$, so its trace is zero and repeated application proves the forward inclusion in (DSP15). Conversely, for the function on its right, each $P^j u \in H^{2m-2j}$, has trace zero for $j < m$, and $P^{j+1} u \in L^2$. By (DSP3), $P^j u \in D(P_D)$, with $P_D(P^j u) = P^{j+1} u$. Reading these identities from $j = m - 1$ back to $j = 0$ gives the recursive domain. ■

5. Compact resolvent

We prove the compactness map needed here directly. Let (u_j) be bounded by M in $H_0^1(X)$. The zero extensions U_j belong to $H^1(\mathbb{R}^n)$ by Section 2 of Boundary energy, local inverses, and harmonic data, have the same derivative-sum norm, and are supported in the fixed compact set $K = \overline{X}$. Cauchy–Schwarz gives

$$|\widehat{U}_j(\xi)| \leq M_0 := |K|^{1/2} M, \quad |\widehat{U}_j(\xi) - \widehat{U}_j(\eta)| \leq M_1 |\xi - \eta|, \quad M_1 := \left(\int_K |x|^2 dx \right)^{1/2} M.$$

The Fourier integral is defined because $U_j \in L^1$; it agrees with the L^2 transform. The second inequality follows from $|e^{-ix \cdot \xi} - e^{-ix \cdot \eta}| \leq |x| |\xi - \eta|$. Plancherel and the weak-derivative identity give the tail bound

$$\int_{\mathbb{R}^n \setminus [-R, R]^n} |\widehat{U}_j(\xi)|^2 d\xi \leq R^{-2} (2\pi)^n M^2.$$

Partition the cube into finitely many cells of side at most δ . On each cell Q , approximate $\widehat{U}_j$ by its value at one fixed point of Q , and set the approximation V_j to zero off the cube. Then

$$\|\widehat{U}_j - V_j\|_2^2 \leq n\delta^2 (2R)^n M_1^2 + R^{-2} (2\pi)^n M^2.$$

For any prescribed positive error choose R first and δ second to make both terms small. The cell values are uniformly bounded by M_0 , so the family (V_j) has finite nets in its finite-dimensional space. Hence $(\widehat{U}_j)$ has finite nets in L^2 . Choosing successively an infinite subsequence in one ball of each finite 2^{-k} -net and then a diagonal subsequence produces an L^2 -Cauchy subsequence. Completeness and Plancherel give an L^2 -convergent subsequence of (U_j) , and restriction followed by

$$\|u_j - u_\ell\|_{L_r^2(X)} \leq r_{\max}^{1/2} \|U_j - U_\ell\|_{L^2(\mathbb{R}^n)}$$

gives convergence in the original weighted space. This proves compactness of $H_0^1(X) \rightarrow L_r^2(X)$, with the support and weight unchanged. By (DSP4) the inverse $K_T = T^{-1}$ therefore is compact on L_r^2 . It is self-adjoint and positive: writing $f = Tu$ gives $(K_T f, f)_r = (u, Tu)_r \geq \|u\|_r^2$. It is injective and has dense range $D(T)$.

Here is the compact positive spectral argument for this exact operator. For any bounded positive self-adjoint operator K , the discriminant of $(K(x + ty), x + ty) \geq 0$, $t \in \mathbb{C}$, proves

$$|(Kx, y)|^2 \leq (Kx, x)(Ky, y).$$

Put $\beta = \sup_{\|x\|=1} (Kx, x)$. Taking the supremum over unit y in this inequality shows $\|Kx\|^2 \leq \beta(Kx, x)$, and consequently $\|K\| = \beta$. If $K \neq 0$, then $\beta > 0$. For unit x_j with $(Kx_j, x_j) \rightarrow \beta$,

$$\|(K - \beta I)x_j\|^2 \leq \beta\{\beta - (Kx_j, x_j)\} \rightarrow 0.$$

When K is compact, a subsequence of Kx_j converges, and the last estimate makes x_j converge along that subsequence to a unit ϕ with $K\phi = \beta\phi$. The orthogonal complement of ϕ is invariant because K is self-adjoint. Repeating the argument on that complement constructs orthonormal eigenvectors with positive nonincreasing eigenvalues β_j , until the restriction becomes zero. Each positive eigenspace is finite-dimensional: an infinite orthonormal sequence in it would have pairwise image distance $\sqrt{2}\beta_j$, contradicting compactness. If the construction is infinite, $\beta_j \rightarrow 0$, by the same compactness argument applied to $K\phi_j = \beta_j\phi_j$. If y is perpendicular to all selected eigenvectors and $Ky \neq 0$, positivity and the displayed Cauchy–Schwarz bound give $(Ky, y) > 0$. But the norm of the restriction after N steps is the next selected β_{N+1} , so $(Ky, y) \leq \beta_{N+1}\|y\|^2 \rightarrow 0$, a contradiction. Thus the only remaining subspace is $\ker K$. For $K = K_T$ that kernel is zero, so the eigenvectors form a complete orthonormal basis. The space $L_r^2(X)$ is infinite-dimensional because a ball inside the nonempty open X contains arbitrarily many pairwise disjoint balls with nonzero indicator functions. Thus the sequence for K_T is infinite. Each $\phi_j = \beta_j^{-1}K_T\phi_j$ belongs to $D(T)$, and $T\phi_j = \beta_j^{-1}\phi_j$. Writing $\tau_j = \beta_j^{-1}$ gives an orthonormal basis ϕ_j , finite multiplicities, and eigenvalues

$$P_D\phi_j = \lambda_j\phi_j, \quad T\phi_j = \tau_j\phi_j, \quad \tau_j = \lambda_j + \mu \geq 1, \quad \tau_j \rightarrow \infty. \quad (\text{DSP16})$$

Indeed the nonzero eigenvalues of T^{-1} are $\tau_j^{-1} > 0$ and tend to zero. There is no kernel, and its range is dense, so these eigenvectors span the whole space. Each eigenfunction belongs to every $D(T^m)$, by its eigenvalue equation. Formula (DSP14) and the Sobolev estimate proved below make it smooth up to the boundary. This is precisely where boundedness of the whole domain yields a global discrete spectrum.

Discrete spectral calculus on the original weighted space

We prove the functional-calculus facts required in the bounded-domain case from this eigenbasis. For $u \in L_r^2(X)$, write $u_j = (u, \phi_j)_r$. Orthogonality gives $\|\sum_{j=1}^N c_j\phi_j\|_r^2 = \sum_{j=1}^N |c_j|^2$. Consequently every ℓ^2 sequence has a convergent series in the complete weighted space. Conversely the coefficients of any u have bounded partial squared sums by orthogonal projection; their series converges, and its difference from u is perpendicular to every ϕ_j . Completeness of the eigenbasis makes that difference zero. Thus the exact maps

$$\mathcal{U} : L_r^2(X) \rightarrow \ell^2, \quad \mathcal{U}u = (u_j)_{j \geq 1}, \quad \mathcal{U}^{-1}(c) = \sum_{j=1}^{\infty} c_j\phi_j \quad (\text{DSP16a})$$

are mutually inverse isometries. Polarization of the finite-sum identity, followed by the norm limits, also proves the inner-product identity.

If $u \in D(T)$, self-adjointness and the eigenvalue equation give $(Tu, \phi_j)_r = (u, T\phi_j)_r = \tau_j u_j$, since τ_j is real. Conversely, if $\sum_j \tau_j^2 |u_j|^2 < \infty$, the finite sums $u^{(N)} = \sum_{j=1}^N u_j\phi_j$ belong to $D(T)$, converge to u , and have $Tu^{(N)}$ convergent to $\sum_j \tau_j u_j\phi_j$. Closedness of T proves that u belongs to its actual domain and has this image. We have proved

$$D(T) = \left\{ u : \sum_j \tau_j^2 |u_j|^2 < \infty \right\}, \quad Tu = \sum_j \tau_j u_j\phi_j. \quad (\text{DSP16b})$$

For each integer $q \geq 1$, induction using $D(T^{q+1}) = \{u \in D(T) : Tu \in D(T^q)\}$ and $\tau_j \geq 1$ gives the exact formulas

$$D(T^q) = \left\{ u : \sum_j \tau_j^{2q} |u_j|^2 < \infty \right\}, \quad T^q u = \sum_j \tau_j^q u_j\phi_j, \quad \|T^q u\|_r^2 = \sum_j \tau_j^{2q} |u_j|^2. \quad (\text{DSP16c})$$

In that induction the higher moment implies every lower moment because $\tau_j \geq 1$, so no recursive domain condition is dropped. The original operator is still $P_D = T - \mu I$; its action in these same coordinates is multiplication by $\lambda_j = \tau_j - \mu$. Its domain agrees with the λ_j^2 moment domain: both inequalities $|\lambda_j| \leq \tau_j + \mu$ and $\tau_j \leq |\lambda_j| + \mu$, together with $(u_j) \in \ell^2$, prove the two inclusions. The same recursive argument for P_D^q uses $|\lambda_j|^{2k} \leq 1 + |\lambda_j|^{2q}$, $0 \leq k \leq q$, and gives its exact $|\lambda_j|^{2q}$ moment domain and action. Multiplication by τ_j^{-q} is bounded on all ℓ^2 , since $\tau_j \geq 1$, and the resulting operator maps into $D(T^q)$. Coordinate multiplication proves both inverse identities with T^q on their respective full and graph domains.

For a Borel set $S \subset \mathbb{R}$, define directly

$$F(S)u = \sum_{j: \lambda_j \in S} u_j \phi_j. \quad (\text{DSP16d})$$

The isometry (DSP16a) proves that this is a bounded orthogonal projection, $F(\mathbb{R}) = I$, and $F(S)F(R) = F(S \cap R)$. For disjoint Borel sets S_k , each coordinate belongs to at most one set. The squared norm of $F(\bigcup_k S_k)u - \sum_{k=1}^N F(S_k)u$ is the sum of $|u_j|^2$ over the remaining coordinates, and tends to zero by convergence of $\sum_j |u_j|^2$. This proves strong countable additivity. Its positive scalar measure is exactly $(F(S)u, u)_r = \sum_{j: \lambda_j \in S} |u_j|^2$. The corresponding cross measure has coefficients $u_j \overline{v_j}$, whose absolute sum is at most $\|u\|_r \|v\|_r$ by Cauchy–Schwarz; it is therefore countably additive as well. These statements construct the projection-valued measure on the original space, with its original density and inner-product convention.

For any finite-valued complex Borel function m on $\mathbb{R}$, put

$$\begin{aligned} D(m(P_D)) &= \left\{ u : \sum_j |m(\lambda_j)|^2 |u_j|^2 < \infty \right\}, \\ m(P_D)u &= \sum_j m(\lambda_j) u_j \phi_j, \\ \|m(P_D)u\|_r^2 &= \sum_j |m(\lambda_j)|^2 |u_j|^2 = \int_{\mathbb{R}} |m(t)|^2 d(F(t)u, u)_r. \end{aligned} \quad (\text{DSP16e})$$

The integral denotes integration against the scalar measure just defined, not differentiation of a function value at an individual point. For a nonnegative simple function its atomic sum formula follows from (DSP16d); increasing simple approximation gives the stated formula for $|m|^2$. Every finite eigenvector sum is in the domain, hence it is dense. The operator is closed: if $u^{(k)} \rightarrow u$ and $m(P_D)u^{(k)} \rightarrow v$, convergence of each coordinate gives $v_j = m(\lambda_j)u_j$; (DSP16a) then puts u in the specified moment domain with image v . Testing the defining adjoint identity against each ϕ_j shows that a vector v in the adjoint domain must satisfy $((m(P_D))^*v)_j = \overline{m(\lambda_j)}v_j$. These coordinates are in ℓ^2 exactly when $v \in D(\overline{m}(P_D))$. Conversely the two convergent series and Cauchy–Schwarz give $(m(P_D)u, v)_r = (u, \overline{m}(P_D)v)_r$ on those domains. Thus $(m(P_D))^* = \overline{m}(P_D)$ with equality of domains. In particular real m give self-adjoint operators. Taking $m(t) = t$ recovers exactly P_D , by (DSP16b) and its shifted-domain comparison, and taking $m(t) = t^q$ recovers its recursive powers proved above.

For bounded m , its domain is the whole space and $\|m(P_D)\| = \sup_j |m(\lambda_j)|$: the upper bound is (DSP16e), and testing ϕ_j gives the reverse bound. Coordinate multiplication proves addition and products of bounded Borel multipliers. More generally, the precise product domain for any two finite-valued Borel functions is

$$\begin{aligned} D(m_1(P_D)m_2(P_D)) &= D(m_2(P_D)) \cap D((m_1m_2)(P_D)), \\ m_1(P_D)m_2(P_D)u &= (m_1m_2)(P_D)u \quad \text{on that domain.} \end{aligned} \quad (\text{DSP16f})$$

Indeed the first factor can act on $m_2(P_D)u$ precisely when both $\sum_j |m_2(\lambda_j)|^2 |u_j|^2$ and $\sum_j |m_1(\lambda_j)m_2(\lambda_j)|^2 |u_j|^2$ are finite. This proves both inclusions in the domain equality. The intersection cannot generally be suppressed: choose $m_2(\lambda_j) = \tau_j$ and $m_1(\lambda_j) = \tau_j^{-1}$, extending both as Borel functions off the discrete eigenvalue set. Their product has the full-space identity domain, whereas the operator product has domain $D(T)$. That domain is proper: select a subsequence $\tau_{j_k} \geq 2^k$ and use coefficients $u_{j_k} = \tau_{j_k}^{-1}$, zero otherwise. Their squared sum is finite but their τ^2 moment is infinite.

Finally $F((-\infty, \lambda])$ is finite rank, because $\lambda_j \rightarrow \infty$. If u lies in its range and $L = \lambda + \mu \geq 1$, every nonzero coordinate has $1 \leq \tau_j \leq L$. Hence (DSP16c) proves directly

$$\|T^q u\|_r^2 \leq L^{2q} \|u\|_r^2 \quad (q \in \mathbb{N}_0). \quad (\text{DSP16g})$$

If the range is zero the same estimate is immediate. This constructs all spectral multipliers and powers needed for the bounded-domain projector arguments without importing the general unbounded spectral theorem. For the arbitrary lower-bounded extension in Section 10, Sections 1–4 of Spectral measures with the original operator domain retained construct the full spectral measure on the unchanged Hilbert space and prove its exact multiplier and power domains.

6. Weighted projector kernels

For the original, unshifted eigenvalues define $E_\lambda = F((-\infty, \lambda]) = 1_{(-\infty, \lambda]}(P_D)$ by the explicit construction (DSP16d)–(DSP16e). The kernel relative to $r(y) dy$ is the finite sum

$$e(x, y, \lambda) = \sum_{\lambda_j \leq \lambda} \phi_j(x) \overline{\phi_j(y)}, \quad E_\lambda f(x) = \int_X e(x, y, \lambda) f(y) r(y) dy. \quad (\text{DSP17})$$

With a linear-first inner product, the coefficient of ϕ_j in this formula is $(f, \phi_j)_r$. Its conjugate is generally different. Weighted orthonormality gives directly

$$\int_X e(x, z, \lambda) e(z, y, \lambda) r(z) dz = e(x, y, \lambda), \quad e(y, x, \lambda) = \overline{e(x, y, \lambda)}. \quad (\text{DSP18})$$

The kernel against Lebesgue measure is $r(y)e(x, y, \lambda)$. The density belongs to the integrated variable y .

7. Parameter Sobolev estimates with the half-space endpoint

Let k, ν be nonnegative integers with $k > \nu + n/2$. For $u \in H^k(X)$ and every $s \geq 1$,

$$s^{k-\nu-n/2} \sum_{|\alpha| \leq \nu} \sup_{\bar{X}} |D^\alpha u|^2 \leq C (\|u\|_{\bar{H}^k(X)}^2 + s^k \|u\|_{L^2(X)}^2). \quad (\text{DSP19})$$

The representative has continuous derivatives through order ν on the closure. The sharper per-derivative half-space assertion, initially for smooth functions compactly supported in the closed half-space, is

$$s^{k-|\alpha|-n/2} \sup_{\mathbb{R}_+^n} |D^\alpha u|^2 \leq C_{k,\alpha} \left(\sum_{|\beta|=k} \|D^\beta u\|_{L^2(\mathbb{R}_+^n)}^2 + s^k \|u\|_{L^2(\mathbb{R}_+^n)}^2 \right), \quad s > 0, \quad (\text{DSP20})$$

where $k > |\alpha| + n/2$. The right side retains only exact-order- k derivatives and the zeroth-order norm. No boundary trace assumption is made in these Sobolev statements.

The one-dimensional estimate, including its boundary term. For a complex-valued smooth function v compactly supported in $[0, \infty)$, direct integration by parts gives

$$\int_0^\infty |v'' + v'|^2 dt = \|v''\|_2^2 - \|v'\|_2^2 + \|v\|_2^2 - |v'(0) + v(0)|^2. \quad (\text{DSP21})$$

In fact $2 \operatorname{Re} \int v'' \overline{v'} = -|v'(0)|^2$, $2 \operatorname{Re} \int v' \overline{v} = -|v(0)|^2$, and $2 \operatorname{Re} \int v'' \overline{v} = -2\|v'\|_2^2 - 2 \operatorname{Re}(v'(0) \overline{v(0)})$. Substitution proves the displayed identity. Its two subtracted squares imply $\|v'\|_2^2 \leq \|v''\|_2^2 + \|v\|_2^2$. Apply this to $v(at)$, $a > 0$: the squared norms scale as $a\|v'\|_2^2$, $a^3\|v''\|_2^2$, and $a^{-1}\|v\|_2^2$. Minimizing $a^2\|v''\|_2^2 + a^{-2}\|v\|_2^2$ proves the exact estimate

$$\|v'\|_2^2 \leq 2\|v''\|_2\|v\|_2. \quad (\text{DSP22})$$

If a norm in that minimum is zero, compact support and $v'' = 0$, or $v = 0$, force $v = 0$, so the same conclusion holds. There is no deletion or absorption of an unknown boundary trace; (DSP21) determines its sign.

It follows, for every integer $k \geq 1$, that

$$\sum_{j=0}^k \|\partial_t^j v\|_2^2 \leq C_k (\|\partial_t^k v\|_2^2 + \|v\|_2^2). \quad (\text{DSP23})$$

Here are the induction details. The assertion for $k = 1$ is an equality with $C_1 = 1$. Assuming it at $k - 1$, apply it to v' to bound $S = \sum_{j=1}^k \|v^{(j)}\|_2^2 \leq C_{k-1} (\|v^{(k)}\|_2^2 + \|v'\|_2^2)$. By (DSP22) and Young's inequality, $\|v'\|_2^2 \leq \varepsilon \|v''\|_2^2 + \varepsilon^{-1} \|v\|_2^2 \leq \varepsilon S + \varepsilon^{-1} \|v\|_2^2$. Choose $\varepsilon = (2C_{k-1})^{-1}$, move $S/2$ to the left, and add $\|v\|_2^2$. Integration in the tangential variables proves the same inequality for the normal derivatives of a half-space function.

An extension preserving the exact highest order. For fixed $k \geq 1$, let $c_1, \dots, c_k$ solve the Vandermonde equations

$$\sum_{j=1}^k c_j (-j)^\ell = 1, \quad 0 \leq \ell < k.$$

Extend $u(z, t)$, $t \geq 0$, by

$$\mathcal{E}_k u(z, t) = \sum_{j=1}^k c_j u(z, -jt), \quad t < 0, \quad \mathcal{E}_k u = u \quad (t \geq 0). \quad (\text{DSP24})$$

The normal derivatives through order $k - 1$ match at zero. Piecewise integration by parts therefore produces no interface delta in derivatives through order k . For every multiindex β of order at most k , the j -th reflected derivative is $c_j (-j)^{\beta_n} (D^\beta u)(z, -jt)$. Substitution $s = -jt$ gives its negative-half-space norm exactly as $|c_j| j^{\beta_n - 1/2} \|D^\beta u\|_{L^2(\mathbb{R}_+^n)}$. Define

$$S_{k, \beta_n} = \sum_{j=1}^k |c_j| j^{\beta_n - 1/2}.$$

The triangle inequality on the negative half-space, followed by adding the squared norm on the positive half-space, proves

$$\|D^\beta \mathcal{E}_k u\|_{L^2(\mathbb{R}^n)} \leq (1 + S_{k,\beta_n}^2)^{1/2} \|D^\beta u\|_{L^2(\mathbb{R}_+^n)}. \quad (\text{DSP25})$$

Thus the normal dilation factors and the finite coefficient sum are both retained. In particular the constant is not merely the Euclidean norm of the coefficient vector.

This retains the derivative order exactly, including order zero. The integer-density and trace construction in Section 2 of Boundary energy, local inverses, and harmonic data extends it to every H^k function. For $k = 0$, extension by zero is sufficient for the norm assertions alone.

For $0 \leq j \leq k$, $|\xi|^{2j} \leq C_k(1 + |\xi|^{2k})$, and the multinomial formula compares $|\xi|^{2k}$ with $\sum_{|\beta|=k} |\xi^\beta|^2$. Plancherel applied to (DSP24) and (DSP25) proves the further exact-order assertion

$$\|u\|_{H^k(\mathbb{R}_+^n)}^2 \leq C_k \left(\sum_{|\beta|=k} \|D^\beta u\|_2^2 + \|u\|_2^2 \right). \quad (\text{DSP26})$$

Thus (DSP23) and the full mixed-derivative estimate are both proved without assigning a zero trace to u .

Fourier proof of the parameter powers. For $U = \mathcal{E}_k u$, Cauchy–Schwarz in Fourier inversion gives

$$|D^\alpha U(x)|^2 \leq C_n \left(\int_{\mathbb{R}^n} \frac{|\xi|^{2|\alpha|}}{|\xi|^{2k} + s^k} d\xi \right) \left((2\pi)^{-n} \int_{\mathbb{R}^n} (|\xi|^{2k} + s^k) |\widehat{U}(\xi)|^2 d\xi \right). \quad (\text{DSP27})$$

The first integral is finite precisely under the stated strict inequality at infinity; it is finite near zero because $s > 0$. The substitution $\xi = s^{1/2} \eta$ makes it $C_{k,\alpha,n} s^{|\alpha|+n/2-k}$. Plancherel, the multinomial comparison, and (DSP25) now give (DSP20). The Fourier integrability also gives continuous derivatives by dominated convergence; approximation extends the assertion from smooth inputs to H^k .

A finite partition of unity, smooth boundary flattening, (DSP24) on boundary patches, and extension by zero on compact interior patches give one extension of a bounded smooth-domain function with

$$\|\mathcal{E}_X u\|_{H^k(\mathbb{R}^n)} \leq C_k \|u\|_{H^k(X)}, \quad \|\mathcal{E}_X u\|_{L^2(\mathbb{R}^n)} \leq C_k \|u\|_{L^2(X)}.$$

Here are the finite-atlas constants behind these two inequalities. Choose boundary charts $\Phi_a : U_a \rightarrow V_a$ and interior charts, and a partition $\{\vartheta_a\}_{a=1}^N$ with $\sum_a \vartheta_a = 1$ on $\overline{X}$, each support compactly contained in its chart. Set

$$j_a^- = \inf_{U_a} |\det D\Phi_a|, \quad j_a^+ = \sup_{U_a} |\det D\Phi_a|, \quad J_k = \max_{a, |\rho| \leq k+1} (\|D^\rho \Phi_a\|_\infty + \|D^\rho \Phi_a^{-1}\|_\infty),$$

and let $Q_k = \max_{a, |\rho| \leq k} \|D^\rho \vartheta_a\|_\infty$. Repeated chain and product rules give, for $|\gamma| \leq k$,

$$\|D^\gamma[(\vartheta_a u) \circ \Phi_a^{-1}]\|_{L^2(V_a)} \leq C(n, k) (j_a^+)^{1/2} J_k^k Q_k \sum_{|\delta| \leq |\gamma|} \|D^\delta u\|_{L^2(U_a)}. \quad (\text{DSP27a})$$

On each flat boundary patch, (DSP25) contributes the factor $(1 + S_{k,\beta_n}^2)^{1/2}$ for each derivative β ; for order zero the factor is $(1 + S_{k,0}^2)^{1/2}$. Pulling back to the original chart uses the inverse chain rule and the Jacobian bound $(j_a^-)^{-1/2}$; pushing forward used $(j_a^+)^{1/2}$ in (DSP27a). The sum over all derivatives through order k and over $a = 1, \dots, N$ is finite. It therefore produces a single constant depending only on

$n, k, N, j_a^\pm, J_k, Q_k$ and the displayed $S_{k,\ell}$, independent of u and s . For density, choose $u_m \in C^\infty(\overline{X})$ with $u_m \rightarrow u$ in $H^k(X)$ (the integer-density statement of Section 2 of Boundary energy, local inverses, and harmonic data), form $\mathcal{E}_X u_m$, and apply (DSP27a) to $u_m - u_\ell$. The extensions are Cauchy in $H^k(\mathbb{R}^n)$, hence converge to $\mathcal{E}_X u$; order-zero convergence uses the order-zero Jacobian, cutoff, and reflection constants, without derivatives of the input. Passing to this limit in (DSP27) proves the parameter estimate for every $u \in H^k(X)$, with the exact $s^{|\alpha|+n/2-k}$ powers unchanged.

The first inequality follows term by term from the product and chain rules; each term has a derivative of u of order at most k and a bounded chart or cutoff coefficient. The second uses only bounded Jacobians and the order-zero instance of (DSP25), so it introduces no derivative of u . Using this extension in (DSP27) proves (DSP19), first with $s^{k-|\alpha|-n/2}$ for each derivative separately. Since $s \geq 1$ and $|\alpha| \leq \nu$, replacing this weight by $s^{k-\nu-n/2}$ and summing proves exactly the displayed bounded-domain statement. The borderline $k = \nu + n/2$ is not part of the assertion. ■

8. Derivatives of the projector kernel

For every pair of multiindices α, β ,

$$|D_x^\alpha D_y^\beta e(x, y, \lambda)| \leq C_{\alpha\beta} \lambda^{(n+|\alpha|+|\beta|)/2}, \quad x, y \in \overline{X}, \quad \lambda \geq 1. \quad (\text{DSP28})$$

Proof. Let $u = E_\lambda f$, and put $L = \lambda + \mu$. Every included T -eigenvalue is in $[1, L]$, so by the proved diagonal estimate (DSP16g) and (DSP14)

$$\|T^q u\|_{L_r^2} \leq L^q \|u\|_{L_r^2}, \quad \|u\|_{H^{2q}} \leq C_q L^q \|u\|_{L_r^2}. \quad (\text{DSP29})$$

Choose any integer q with $2q > |\alpha| + n/2$, and apply the per-derivative form of (DSP19) with $k = 2q$ and $s = L$. This gives

$$\sup_{\overline{X}} |D^\alpha u| \leq C_\alpha L^{n/4+|\alpha|/2} \|u\|_{L_r^2}. \quad (\text{DSP30})$$

Only even integer Sobolev orders were required, and there is no additional ε loss in the spectral exponent.

For fixed x , the linear functional $f \mapsto \partial_x^\alpha E_\lambda f(x)$ on L_r^2 has squared norm

$$\sum_{\lambda_j \leq \lambda} |\partial^\alpha \phi_j(x)|^2 \leq C_\alpha L^{n/2+|\alpha|}. \quad (\text{DSP31})$$

This follows either by its finite-dimensional Riesz representative, or by choosing coefficients proportional to the conjugates of the displayed derivative values in (DSP30); the zero row is immediate. Cauchy–Schwarz in the finite kernel sum yields

$$|\partial_x^\alpha \partial_y^\beta e| \leq \left(\sum |\partial^\alpha \phi_j(x)|^2 \right)^{1/2} \left(\sum |\partial^\beta \phi_j(y)|^2 \right)^{1/2} \leq C_{\alpha\beta} L^{(n+|\alpha|+|\beta|)/2}.$$

Replacing ordinary derivatives by D multiplies the expression by a constant of absolute value one. Since $\lambda \geq 1$ and μ is fixed, $L \leq (1 + \mu)\lambda$; this proves (DSP28) for the original unshifted parameter. For a joint multiindex $\gamma = (\alpha, \beta)$, the exponent is exactly $(n + |\gamma|)/2$. ■

9. Counting bound

The projector is finite rank, so its trace and diagonal integral are

$$N(\lambda) = \text{Tr } E_\lambda = \sum_{\lambda_j \leq \lambda} 1 = \int_X e(x, x, \lambda) r(x) dx \leq C\lambda^{n/2}, \quad \lambda \geq 1. \quad (\text{DSP32})$$

The middle equality follows by integrating the finite sum (DSP17) and using $\|\phi_j\|_{L^2_r} = 1$. The bound follows from (DSP28) with both derivatives zero and the finite weighted volume of X . Multiplicities are included. No lower asymptotic or leading Weyl coefficient is asserted.

10. Arbitrary lower-bounded extensions

For this section X is an arbitrary open subset of $\mathbb{R}^n$; it need not be bounded. The same expression (DSP2) has smooth coefficients and smooth positive density on X , real symmetric positive definite principal matrix at every point, and is symmetric on $C_c^\infty(X)$ in $H = L^2(X, r dx)$. Only local coefficient bounds are required. Let A be any self-adjoint extension of this test-domain operator with a finite lower bound a , so $A \geq aI$.

Let ω be a relatively open smooth portion of the boundary such that X locally lies on one side of it and the coefficients and positive density extend smoothly across it, retaining positive definiteness. When boundary conclusions are used, assume

$$\{v \in C_c^\infty(X \cup \omega) : v|_\omega = 0\} \subset D(A). \quad (\text{DSP33})$$

Here a function smooth on $X \cup \omega$ means a one-sided restriction smooth across the patch, and its compact support stays away from all other boundary portions. This is an inclusion of test functions in the actual operator domain, not an assumption that arbitrary domain elements already have a trace. We now prove their local trace and regularity.

For every $\lambda \in \mathbb{R}$, $E_\lambda = 1_{(-\infty, \lambda]}(A)$ has a smooth kernel $e_A(x, y, \lambda)$ locally on $X \times X$. Under (DSP33) it is smooth up to the allowed boundary in both variables, and for every compact $K \subset X \cup \omega$ and $\lambda \geq 1$,

$$|D_x^\alpha D_y^\beta e_A(x, y, \lambda)| \leq C_{\alpha\beta, K} \lambda^{(n+|\alpha|+|\beta|)/2}, \quad x, y \in K. \quad (\text{DSP34})$$

For an arbitrary compact subset of $(X \cup \omega)^2$, use the union of its two compact projections as K . The kernel is taken against the original measure $r(y) dy$.

Local regularity of the actual operator domain

For $u \in D(A)$, interior testing and symmetry give $Pu = Au$ distributionally: for $v \in C_c^\infty(X)$, $(u, Pv)_r = (Au, v)_r$. Under (DSP33) this identity holds for zero-trace smooth tests up to the boundary as well. We first prove, on each compact allowed patch,

$$\|u\|_{H^2(K)} \leq C_K (\|Au\|_H + \|u\|_H), \quad \gamma u = 0 \text{ on } K \cap \omega. \quad (\text{DSP35})$$

The notation $H^s(K)$ in local estimates means restriction to a fixed finite collection of smaller coordinate neighborhoods covering K ; the norm is the sum of their norms after cutoffs equal to one near K .

Choose a bounded smooth auxiliary domain $\Omega \subset X$, whose boundary agrees with the physical smooth boundary in a neighborhood of the patch, and whose other boundary stays away from a real smooth cutoff χ . Let $\chi = 1$ on a smaller neighborhood of K , with support in the selected chart and away from the artificial part of $\partial\Omega$. For an interior patch choose $\Omega \Subset X$. Here is an explicit boundary construction in flat coordinates (z, t) , with $t > 0$ inside. Choose $c, d > 0$ and a smooth function $\psi(s)$ equal to zero for $s \leq d^2$, strictly increasing thereafter, and exceeding c^2 before the edge of the chart. The region

$$\Omega_{\text{flat}} = \{(z, t) : t^2 - 2ct + \psi(|z|^2) < 0\}$$

is bounded, lies in $t > 0$, and has bottom boundary $t = 0$ when $|z| < d$. Its defining function has nonzero gradient on its zero set: away from $t = c$ use its t -derivative, and at $t = c$ use $\psi'(|z|^2) > 0$ and $|z| > d$. Take the support of χ in $|z| < d$, $0 \leq t < c$, and scale this construction to fit the chart before pulling it back. For $n = 1$, use the interval $(0, 2c)$. All coefficients on the resulting compact closure are smooth with positive ellipticity and density bounds.

For the boundary construction just used, strict increase alone does not imply that the derivative is positive at the level defining the artificial boundary. Use the following explicit choice, with the same original constants and domain. Choose $s_e > d^2$ inside the tangential chart and put

$$I_e = \int_{d^2}^{s_e} \exp\left(-\frac{1}{v-d^2}\right) dv > 0, \quad L > \frac{c^2}{I_e}, \quad \psi(s) = \begin{cases} 0, & s \leq d^2, \\ L \int_{d^2}^s \exp\left(-\frac{1}{v-d^2}\right) dv, & s > d^2. \end{cases} \quad (\text{DSP-AUX1})$$

The integral at its lower endpoint is the integral of the function with value zero there. Every derivative of $\exp(-1/w)$ for $w > 0$ is that exponential times a polynomial in w^{-1} ; this follows by induction using the product rule. Each such product tends to zero as $w \downarrow 0$: after putting $y = w^{-1}$, the estimate $e^y \geq y^m/m!$, with m larger than the polynomial degree, proves the limit. Extending this function by zero to $w \leq 0$ is therefore smooth, and integrating it proves that the displayed ψ is smooth across d^2 . Moreover $\psi'(s) = L \exp(-1/(s-d^2)) > 0$ for every $s > d^2$, and $\psi(s_e) = LI_e > c^2$. For $s \geq d^2 + 1$, its derivative is at least L/e , so it tends to infinity.

The original defining function is $F(z, t) = t^2 - 2ct + \psi(|z|^2)$. On its negative set, $(t-c)^2 < c^2 - \psi(|z|^2) \leq c^2$, so $0 < t < 2c$ and $|z|^2 < s_e$. Choose the original positive constants and chart radii so that this bounded cylinder fits in the chart. When $|z| < d$, the bottom boundary remains exactly $t = 0$. On the zero set with $t \neq c$, the derivative $\partial_t F = 2(t-c)$ is nonzero. At $t = c$, the zero-set equation is $\psi(|z|^2) = c^2 > 0$, so $|z| > d$ and $\nabla_z F = 2\psi'(|z|^2)z \neq 0$.

Here are explicit smooth boundary graphs, so this gradient test requires no unproved implicit-function assertion. Near a point with $t \neq c$, keep the sign of $t - c$; the zero set is the graph $t = c + \text{sgn}(t - c)\sqrt{c^2 - \psi(|z|^2)}$, whose radicand is positive near that point. Near a point with $t = c$, let h be the inverse of ψ on a small interval about $|z|^2 > d^2$. Its derivative has a positive lower bound there. The mean-value theorem bounds the inverse difference quotient and proves continuity of h ; applying that theorem once more to $\psi(h(y + \varepsilon)) - \psi(h(y)) = \varepsilon$ gives $h'(y) = 1/\psi'(h(y))$. Differentiating this identity inductively proves that h is smooth. Choose a nonzero coordinate z_i of the point and keep its sign. The zero set is then the graph

$$z_i = \text{sgn}(z_i) \sqrt{h(c^2 - (t-c)^2) - \sum_{j \neq i} z_j^2},$$

whose radicand is $z_i^2 > 0$ at the point and stays positive nearby. These graphs cover the zero set and prove smoothness of the entire auxiliary boundary with the required physical flat patch. The nonzero

defining derivative on each graph also shows that the domain occupies one side. Pullback keeps the previously specified domain, coefficients, density, cutoffs and transposition identities. In dimension one the previously specified interval $(0, 2c)$ supplies the construction.

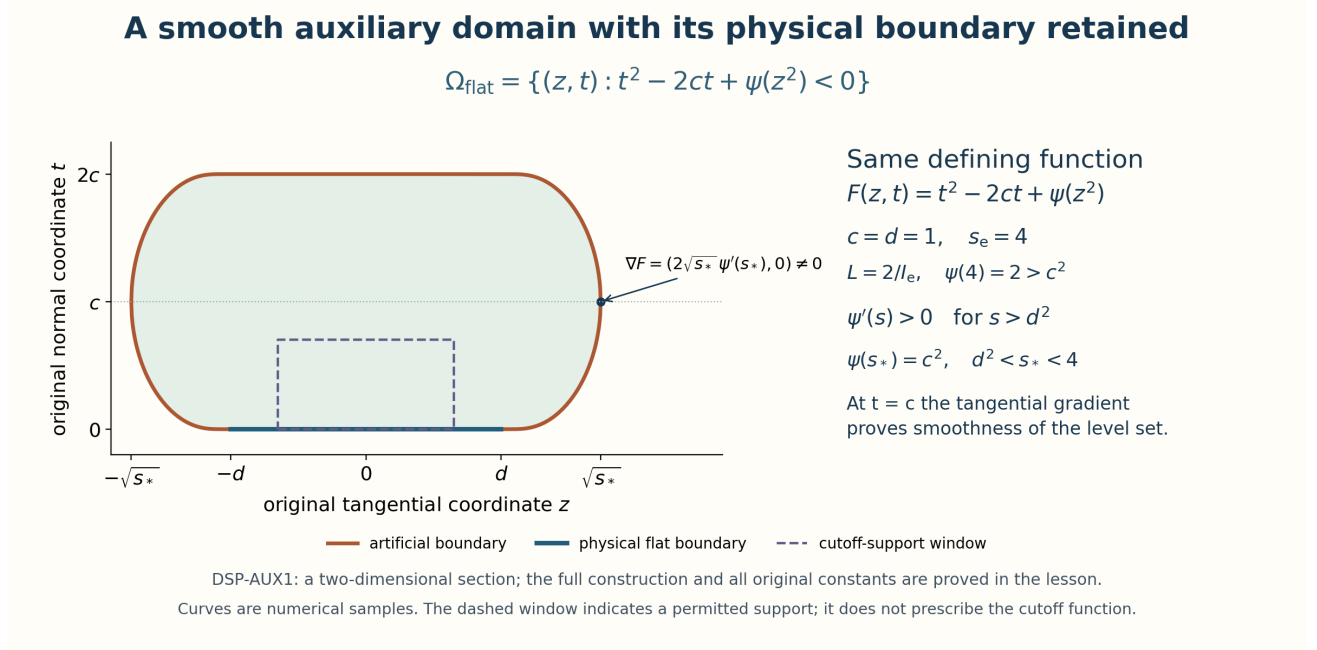


Figure 2: A two-dimensional section of the explicit auxiliary-domain construction (DSP-AUX1), with the sample constants $c = d = 1$, $s_e = 4$, and $L = 2/I_e$. The original domain is $t^2 - 2ct + \psi(z^2) < 0$. The physical flat boundary, artificial boundary, cutoff-support window and nonzero level gradient are distinguished. The curve is a numerical sample of the exact displayed formula; s_* is the unique solution of $\psi(s_*) = c^2$, not a substituted coordinate.

Choose a local shift σ making the Dirichlet form $q_{\Omega, \sigma}$, defined by (DSP5) on Ω , coercive. For $v \in H_0^1(\Omega)$ define the conjugate-linear functional

$$F(v) = (\chi(Au + \sigma u), v)_{r, \Omega} - (u, [P, \chi]v)_{r, \Omega}. \quad (\text{DSP36})$$

The commutator $[P, \chi]$ has order one and smooth bounded coefficients on its compact support. Thus

$$|F(v)| \leq C(\|Au\|_H + \|u\|_H)\|v\|_{H^1(\Omega)}.$$

The energy construction of Section 1 produces $w \in H_0^1(\Omega)$ with $q_{\Omega, \sigma}(w, v) = F(v)$ and

$$\|w\|_{H^1(\Omega)} \leq C(\|Au\|_H + \|u\|_H). \quad (\text{DSP37})$$

For a smooth zero-trace v on $\overline{\Omega}$, the function χv , extended by zero past the artificial boundary, belongs to the test class (DSP33), or is a compact interior test. The domain identity and the product rule give

$$(\chi u, (P + \sigma)v)_{r, \Omega} = (Au, \chi v)_{r, \Omega} + \sigma(u, \chi v)_{r, \Omega} - (u, [P, \chi]v)_{r, \Omega} = F(v). \quad (\text{DSP38})$$

Also $q_{\Omega,\sigma}(w, v) = (w, (P + \sigma)v)_{r,\Omega}$. To verify this last equality for $w \in H_0^1$, approximate w in H^1 by compact interior tests, use formal symmetry there, and pass to the limit. Smooth zero-trace functions are dense in $H^2(\Omega) \cap H_0^1(\Omega)$ in H^2 : in a flattened boundary chart use the odd H^2 extension, mollify by an even mollifier, and use cutoffs; Section 2 of Boundary energy, local inverses, and harmonic data and the density proof after Section 4 of Boundary energy, local inverses, and harmonic data establish the trace and interface cancellation. Interior charts use ordinary convolution. Consequently (DSP38) and the corresponding equality for w hold for every such H^2 test.

The coefficient identities used in this auxiliary problem are invariant under the flattening. If Φ is the chart, put

$$\tilde{r} = (r \circ \Phi^{-1})|\det D\Phi^{-1}|, \quad \tilde{g}^{ab} = (g^{jk} \circ \Phi^{-1})(\partial_j \Phi_a \circ \Phi^{-1})(\partial_k \Phi_b \circ \Phi^{-1}),$$

and let $\tilde{b}, \tilde{c}$ be defined by the full chain rule in the unweighted divergence-form identity

$$\tilde{P}v = -\partial_a(\tilde{g}^{ab}\partial_b v) + \tilde{b}^b\partial_b v + \tilde{c}v, \quad \tilde{P}v = (P(v \circ \Phi)) \circ \Phi^{-1}.$$

Thus $\tilde{b}$ is the coefficient before rewriting the expression relative to the transformed density (it includes every first-order term created by the coordinate change and by expanding the principal divergence). With $\tilde{a}^b = \tilde{b}^b + \tilde{r}^{-1}\partial_a(\tilde{r})\tilde{g}^{ab}$, formal symmetry on compact tests gives the pointwise identities

$$\tilde{a}^b = -\overline{\tilde{a}^b}, \quad \tilde{c} - \overline{\tilde{c}} = \tilde{r}^{-1}\partial_b(\tilde{r}\tilde{a}^b). \quad (\text{DSP37a})$$

The Jacobian is already included in $\tilde{r}$, so these are exactly the pullbacks of the unflattened identities. Integration by parts using (DSP37a) proves symmetry of $q_{\Omega,\sigma}$ on $H_0^1(\Omega)$. Moreover, for $v \in H^2(\Omega) \cap H_0^1(\Omega)$, bounded transformed coefficients give

To specify the constant, all norms in the following sums are taken on the fixed auxiliary coordinate domain. Put

$$G_2 = \sum_{a,b=1}^n \|\tilde{g}^{ab}\|_\infty, \quad G_1 = \sum_{b=1}^n \left(\|\tilde{b}^b\|_\infty + \sum_{a=1}^n \|\partial_a \tilde{g}^{ab}\|_\infty \right), \quad C_0 = \|\tilde{c}\|_\infty + |\sigma|.$$

The unweighted divergence-form definition in (DSP37a) expands to

$$(\tilde{P} + \sigma)v = -\sum_{a,b} \tilde{g}^{ab}\partial_a\partial_b v + \sum_b \left(\tilde{b}^b - \sum_a \partial_a \tilde{g}^{ab} \right) \partial_b v + (\tilde{c} + \sigma)v.$$

Every derivative norm displayed here is bounded by the inhomogeneous H^2 norm (DSP6), and $\|f\|_{L_{\tilde{r}}^2} \leq \tilde{r}_{\max}^{1/2}\|f\|_2$. The triangle inequality therefore proves the explicit bound

$$\|(\tilde{P} + \sigma)v\|_{L_{\tilde{r}}^2(\Omega)} \leq \tilde{r}_{\max}^{1/2}(G_2 + G_1 + C_0)\|v\|_{H^2(\Omega)}. \quad (\text{DSP37b})$$

The transformed coefficient used here is $\tilde{b}^b = \tilde{a}^b - \sum_a (\partial_a \log \tilde{r})\tilde{g}^{ab}$. Consequently, if the constant is instead expressed using $\tilde{a}$, it also depends on $\|\partial_a \log \tilde{r}\|_\infty$. Minimum and maximum density bounds alone do not bound that dependence. These coefficient norms are finite on the fixed compact closure, as required by the auxiliary problem.

The coercive solution for every $f \in L_{\tilde{r}}^2(\Omega)$ lies in $H_0^1(\Omega)$, and Section 6 of Boundary energy, local inverses, and harmonic data (whose hypotheses are verified by (DSP37a)) upgrades it to $H^2(\Omega)$. Thus

$P_{D,\Omega} + \sigma : H^2(\Omega) \cap H_0^1(\Omega) \rightarrow L_r^2(\Omega)$ is both continuous by (DSP37b) and onto. The fixed chart norm equivalences $\tilde{r}_{\min}^{1/2} \|v\|_{L^2} \leq \|v\|_{L_r^2} \leq \tilde{r}_{\max}^{1/2} \|v\|_{L^2}$ and the analogous pullback H^2 bounds transfer this statement back to $L_r^2(\Omega)$, which is the notation used in (DSP36)–(DSP38).

The shifted Dirichlet map $P_{D,\Omega} + \sigma : H^2(\Omega) \cap H_0^1(\Omega) \rightarrow L_r^2(\Omega)$ is onto: Section 1 solves for each L^2 datum in H_0^1 , and Section 3 at $k = 0$ places that solution in H^2 . Therefore $(\chi u - w, f)_{r,\Omega} = 0$ for every $f \in L_r^2(\Omega)$, proving $\chi u = w$. This proves local H^1 , zero trace, and their bound before any trace of u was invoked.

On a slightly smaller patch the equation is $Pu = Au \in L^2$, and the already verified H^1 and zero trace allow Section 6 of Boundary energy, local inverses, and harmonic data to give (DSP35). Interior patches use the interior regularity theorem after this same transposition step. No multiplication-invariance of the unknown domain $D(A)$ is required; only the explicit smooth functions χv belong to it.

Iterated graph estimates and bounded spectral subspaces

For each integer $q \geq 1$, sufficiently many nested allowed patches give

$$\|u\|_{H^{2q}(K)} \leq C_{q,K} \sum_{j=0}^q \|A^j u\|_H, \quad u \in D(A^q). \quad (\text{DSP39})$$

To prove this, for $q = 1$ use (DSP35). Suppose the result holds for $q - 1$. Then $Au \in D(A^{q-1})$, so on a slightly larger patch it belongs to H^{2q-2} , bounded by $C \sum_{j=1}^q \|A^j u\|_H$. The function u has local H^1 and zero trace by (DSP35). The smooth bootstrap of Section 3, or exactly Section 7 of Boundary energy, local inverses, and harmonic data on these nested patches, applied to $Pu = Au$ now gives local H^{2q} with the bound (DSP39). The interior differentiated-equation proof has the same estimates without the trace. This induction explicitly controls the localization commutators by the previous levels on larger patches.

Choose once and for all $\rho \geq 1 + \max(0, -a)$, so $B = A + \rho I \geq I$. Section 4 of Spectral measures with the original operator domain retained constructs the bounded positive inverse $S = B^{-1}$ on this original weighted Hilbert space. For its PVM E , (LB17) gives exactly $E_\lambda = E([(\lambda + \rho)^{-1}, (a + \rho)^{-1}])$ when $\lambda \geq a$; both endpoints are included. If $\lambda < a$, then $E_\lambda = 0$. Otherwise $u \in E_\lambda H$ lies in every recursively defined $D(A^q)$ by (LB16), and on its spectral support $1 \leq t + \rho \leq L = \lambda + \rho$. For $\lambda \geq 1$, (DSP1) and (LB18)–(LB19) give

$$\|B^j u\|_H \leq L^j \|u\|_H, \quad \sum_{j=0}^q \|A^j u\|_H \leq C_{q,\rho} L^q \|u\|_H. \quad (\text{DSP40})$$

One explicit permitted constant is $C_{q,\rho} = (q+1)(1+\rho)^q$. Indeed $|t| \leq L + \rho \leq (1+\rho)L$ on the selected interval, and each of its $q+1$ moments is bounded by that common order- q bound because $L \geq 1$. The companion proves the equality of both recursive power conventions, including the recurrence through Au used in (DSP39).

Choose a smooth cutoff χ equal to one near K . Apply (DSP39) to u on a neighborhood of its support, and the product rule then bounds χu in H^{2q} ; no membership of χu in $D(A)$ is asserted. Use (DSP19), or its chart proof, on χu with $s = L$ and an even order $2q > |\alpha| + n/2$. Its L^2 norm is bounded by $C_K \|u\|_H$ using local density comparison, and its H^{2q} norm by $C_{q,K} L^q \|u\|_H$ from (DSP39). Thus

$$\sup_K |\partial^\alpha u| \leq C_{\alpha,K} L^{n/4+|\alpha|/2} \|u\|_H, \quad u \in E_\lambda H. \quad (\text{DSP41})$$

For any fixed finite $\lambda \geq a$, the same argument with $s = \max(1, \lambda + \rho)$ proves smoothness and finite evaluation norms. Equation (DSP41) records the uniform high-energy dependence needed for (DSP34).

Kernel construction with the sharp endpoint retained

For each x in an allowed patch, the bounded linear functional $f \mapsto E_\lambda f(x)$ has a unique representative $h_x \in H$ with

$$E_\lambda f(x) = (f, h_x)_r. \quad (\text{DSP42})$$

It vanishes on $(I - E_\lambda)H$, so $h_x \in E_\lambda H$. For every multiindex α , the analogous functional $f \mapsto \partial^\alpha E_\lambda f(x)$ has a representative $h_{x,\alpha} \in E_\lambda H$, and (DSP41) gives

$$\|h_{x,\alpha}\|_H \leq C_{\alpha,K} L^{n/4+|\alpha|/2}. \quad (\text{DSP43})$$

These are the actual Hilbert-norm derivatives of $x \mapsto h_x$. Here is the operator-norm argument, including the boundary. On a coordinate neighborhood U with compact closure in the allowed chart, and for $m = 1, 2$, define $M_{\alpha,m}$ to be the supremum, over $x \in \bar{U}$, $\|f\|_H \leq 1$, and real vectors $|v_1|, \dots, |v_m| \leq 1$, of

$$\left| \sum_{i_1, \dots, i_m=1}^n (v_1)_{i_1} \cdots (v_m)_{i_m} \partial^{\alpha+e_{i_1}+\dots+e_{i_m}} E_\lambda f(x) \right|.$$

These are directional tensor norms. They are finite by (DSP41): each is at most $n^{m/2}$ times the largest componentwise bound of order $|\alpha| + m$, since $\|v_j\|_{\ell^1} \leq \sqrt{n} |v_j|$. This retains the dimension factors needed for the Taylor remainder. If $x, x + \delta \in \bar{U}$ and the segment $x + t\delta$ stays in $\bar{U}$, Taylor's integral formula gives, uniformly for $\|f\|_H \leq 1$,

$$\left| \partial^\alpha E_\lambda f(x + \delta) - \partial^\alpha E_\lambda f(x) - \sum_{i=1}^n \delta_i \partial^{\alpha+e_i} E_\lambda f(x) \right| \leq \frac{1}{2} M_{\alpha,2} |\delta|^2. \quad (\text{DSP43a})$$

Taking the supremum over the unit ball of H identifies (DSP43a) with

$$\left\| h_{x+\delta,\alpha} - h_{x,\alpha} - \sum_i \delta_i h_{x,\alpha+e_i} \right\|_H \leq \frac{1}{2} M_{\alpha,2} |\delta|^2.$$

Hence every first difference quotient converges in H , with an $O(|\delta|)$ remainder, and iteration with derivatives of order $|\alpha| + m$ proves C^∞ Hilbert-norm differentiability in the interior. At a curved boundary an ambient straight segment between two allowed points need not stay in the domain. We therefore apply the remainder argument in the actual flattened chart, retaining every derivative of that chart. Let $\Psi = \Phi^{-1}$ send a convex coordinate half-ball V_+ into the allowed domain, and work on a smaller closed half-ball. For a fixed ambient multiindex α , put

$$k_\alpha(y) = h_{\Psi(y),\alpha}, \quad \ell_\alpha(y; f) = \partial_x^\alpha E_\lambda f(\Psi(y)). \quad (\text{DSP43b})$$

Thus $\ell_\alpha(y; f) = (f, k_\alpha(y))_r$. Repeated chart derivatives keep their full coefficients. Starting with $c_{0,\gamma}^\alpha = \delta_{\alpha\gamma}$, define along any fixed ordered sequence of differentiations

$$\begin{aligned} c_{\beta+e_j,\gamma}^\alpha &= \partial_{y_j} c_{\beta,\gamma}^\alpha + \sum_{l=1}^n c_{\beta,\gamma-e_l}^\alpha \partial_{y_j} \Psi_l, \\ \partial_y^\beta \ell_\alpha(y; f) &= \sum_\gamma c_{\beta,\gamma}^\alpha(y) \partial_x^\gamma E_\lambda f(\Psi(y)). \end{aligned} \quad (\text{DSP43c})$$

Absent coefficients are zero. These finite sums have $|\gamma| \leq |\alpha| + |\beta|$. The ordinary product and chain rules prove the recursion, so all coefficients are real and smooth; no coordinate factor is absorbed into the evaluation vector. Equation (DSP41) bounds every displayed derivative, uniformly over $\|f\|_H \leq 1$ on the compact chart. In particular let $\widetilde{M}_{\alpha,m}$, for $m = 1, 2$, be the supremum of its order- m directional tensor norm in y , with real unit directions, over that same unit ball and closed half-ball. It is finite: the componentwise bound from (DSP43c), multiplied by $n^{m/2}$, is an explicit sufficient bound.

All segments between points in this half-ball stay in it. The scalar Taylor formula in these coordinates, followed by the supremum over $\|f\|_H \leq 1$, gives

$$\left\| k_\alpha(y + \delta) - k_\alpha(y) - \sum_{j,l} \delta_j (\partial_{y_j} \Psi_l)(y) h_{\Psi(y), \alpha + e_l} \right\|_H \leq \frac{1}{2} \widetilde{M}_{\alpha,2} |\delta|^2, \quad (\text{DSP43d})$$

$$\|k_\alpha(y + \delta) - k_\alpha(y)\|_H \leq \widetilde{M}_{\alpha,1} |\delta|.$$

The second bound gives an approach-independent Hilbert limit at the flat wall. That limit is the representative of the actual boundary functional: for every f , the smooth one-sided representative of $E_\lambda f$ has its scalar derivative limit there, and taking the Hilbert pairing commutes with norm convergence. Formula (DSP43d) gives all first tangential and inward derivatives at the wall. Applying the same argument to every α , or the complete recursion (DSP43c), gives all higher Hilbert-norm derivatives. In particular, the return to the original coordinates is exact:

$$\partial_{y_j} k_\alpha = \sum_l (\partial_{y_j} \Psi_l) h_{\Psi(y), \alpha + e_l}, \quad \sum_j (\partial_{x_i} \Phi_j)(x) \partial_{y_j} k_\alpha(\Phi(x)) = h_{x, \alpha + e_i}. \quad (\text{DSP43e})$$

The second identity follows from $D\Psi(\Phi(x))D\Phi(x) = I$, including all entries and their order. It holds in the interior and extends to the wall by the continuous jets just constructed. Constants in an ambient distance comparison include the Lipschitz constant of Φ ; the original ambient tensor constant is not silently reused for a curved path. This proves smoothness up to the allowed boundary in the original coordinates.

For each desired finite derivative order one can also multiply by a fixed compact chart cutoff and apply (DSP24) to this Hilbert-valued map at an integer order M above that derivative order plus $n/2$. The same interface cancellations and triangle inequalities apply in the Hilbert norm, giving an H^M extension; its Fourier Cauchy–Schwarz estimate gives the claimed finite-order continuous derivatives. This is a further verification of the one-sided jets, without requiring one finite reflection operator to extend every order simultaneously. The joint kernel below is obtained from the two actual Hilbert-smooth factors.

Since $h_x \in E_\lambda H$, its local representative satisfies $h_x(y) = (h_x, h_y)_r$. Define

$$e_A(x, y, \lambda) = \overline{h_x(y)} = (h_y, h_x)_r. \quad (\text{DSP44})$$

The two Hilbert-norm-smooth factors prove joint smoothness. They also prove

$$\partial_x^\alpha \partial_y^\beta e_A(x, y, \lambda) = (h_{y,\beta}, h_{x,\alpha})_r,$$

whose absolute value is bounded by the product in (DSP43). As $L = \lambda + \rho \leq (1 + \rho)\lambda$ for $\lambda \geq 1$, this is precisely (DSP34), including $D = -i\partial$. Equation (DSP42) gives the kernel representation against $r(y)dy$; for general $f \in H$, the integral is the L^2 pairing, and for compactly supported data it is the ordinary locally smooth-kernel integral. The endpoint is exactly $(-\infty, \lambda]$, since that was the projector in every step. No boundary limit of a resolvent or approximation of a discontinuous multiplier is involved.

A related consequence now follows with its full multiplier class: if m is any bounded Borel function whose spectral support is bounded above, in the exact sense

$$F_A(\{t > b : m(t) \neq 0\}) = 0 \quad (\text{DSP45a})$$

for some finite real b , then $m(A) = E_b m(A) E_b$ has the local smooth kernel

$$e_m(x, y) = (m(A) h_y^{(b)}, h_x^{(b)})_r. \quad (\text{DSP45})$$

Here take $b \geq a$; if the original bound is less than a , the multiplier is zero by the lower spectral concentration, and its kernel is zero. Bounded multiplicativity proves $m(A) = E_b m(A) E_b$: the two discarded products have multiplier $m 1_{(b, \infty)}$, whose squared norm on every vector is zero by (DSP45a) and (LB14). Conversely, let $C_k = \{t > b : |m(t)| \geq 1/k\}$. The compression identity gives $m(A) F_A(C_k) = 0$, while (LB14) gives

$$0 = \|m(A) F_A(C_k) u\|_H^2 = \int_{C_k} |m(t)|^2 d(F_A(t) u, u) \geq k^{-2} \|F_A(C_k) u\|_H^2. \quad (\text{DSP45c})$$

Thus $F_A(C_k) = 0$ for every k . These sets increase to $\{t > b : m(t) \neq 0\}$, so strong countable additivity gives (DSP45a). The hypothesis therefore records exactly the support needed for this compression.

Its adjoint evaluation vector is $m(A)^* h_x^{(b)}$; evaluating that vector at y and conjugating with the original linear-first convention proves (DSP45). Differentiation and Cauchy–Schwarz bound each mixed derivative by $\|m\|_{F_A, \infty} \|h_{y, \beta}^{(b)}\| \|h_{x, \alpha}^{(b)}\|$, where the essential bound is relative to this projection-valued measure. In particular for $b \geq \max(1, a)$,

$$|\partial_x^\alpha \partial_y^\beta e_m(x, y)| \leq C_{\alpha\beta, K} \|m\|_{F_A, \infty} (b + \rho)^{(n+|\alpha|+|\beta|)/2}. \quad (\text{DSP45b})$$

For every other finite $b \geq a$, choose the evaluation vectors at $\max(1, b)$ instead; the same compression identity and derivative argument give a finite smooth kernel. Smooth compactly supported multipliers and the discontinuous function $1_{(-\infty, b]}$ are both included. No compact topological support, smoothness of m , global finite rank, or finite global trace follows from or is needed by this argument. Also (DSP39) directly proves $\chi(A + \rho)^{-q} : H \rightarrow H^{2q}$ locally: $\|A^j (A + \rho)^{-q}\| \leq \sup_{t \geq a} |t|^j / (t + \rho)^q < \infty$ for $0 \leq j \leq q$. These facts do not require or assert a bound on resolvents uniformly across real spectral poles.

The construction used no compact embedding. It therefore proves local sharp-projector estimates for an arbitrary lower-bounded extension, while leaving global rank and trace to their actual hypotheses. If (DSP33) is absent, the interior conclusion still holds; the proof makes no claim of zero trace or of these Dirichlet boundary estimates at that boundary. ■

Exercises and solutions

Exercise 1. Density and conjugation. Compute $E_\lambda(E_\lambda f)$ using both the eigenbasis and the weighted kernel, and determine the coefficient of one eigenfunction when the inner product is linear first.

Solution. Formula (DSP17) gives $\int e(x, y, \lambda) f(y) r(y) dy = \sum_{\lambda_j \leq \lambda} \phi_j(x) (f, \phi_j)_r$. Applying the projector again retains each of these same coefficients. In the double integral, the middle integration gives $\int \phi_j(z) \phi_k(z) r(z) dz = \delta_{jk}$, so it reduces to the original single integral. If the integration measure is dy , its kernel must be $r(y) e(x, y, \lambda)$. For $f = i\phi_j$, the correct coefficient is i , whereas $(\phi_j, f)_r = -i$, explicitly detecting the reversed-inner-product error.

Exercise 2. The sign and coefficient requirements. Expand $[D_j, P]$ in divergence coordinates, solve for $D_n^2 u$, and give sufficient coefficient bounds for the k -th boundary estimate.

Solution. Since D_j commutes with each ordinary derivative, the product rule gives exactly (DSP13). Expanding P gives (DSP10), because $-\partial_j(g^{jk}\partial_k u) = D_j(g^{jk}D_k u)$ and $b^k\partial_k = ib^k D_k$. Hence the coefficient of Pu in (DSP11) is $+(g^{nn})^{-1}$. A sufficient finite collection for (DSP8) consists of a positive lower ellipticity bound, positive lower and upper density bounds, bounded derivatives of g, r through order $k+1$ and of b, c through order k , and bounded chart, inverse-chart, and cutoff derivatives through order $k+2$ on the chosen patches. More derivatives may be bounded without harm; smoothness supplies them all. In (DSP12) one derivative of a divergence coefficient may already have been used in expanding the first-order term, which explains why a C^k bound on g alone is not the asserted sufficient collection. The transformed coefficients must be measured after their actual chart changes.

Exercise 3. A concrete failure of compactness. On $X = \{x_n > 0\}$, take $r = 1$, $P = -\Delta$, and the Dirichlet realization. Show that its resolvent is not compact, and identify the local kernel statements that survive.

Solution. Odd extension in x_n , divided by $\sqrt{2}$, is a unitary map from $L^2(X)$ to the closed subspace of odd $L^2(\mathbb{R}^n)$ functions. The Fourier realization of $-\Delta$, multiplication by $|\xi|^2$, preserves this subspace and induces the self-adjoint nonnegative Dirichlet operator. The odd-extension and zero-trace statements in Sections 2–4 of Boundary energy, local inverses, and harmonic data identify its domain with $H^2(X) \cap H_0^1(X)$.

Choose a nonzero smooth function w supported in a ball of radius $1/4$ centered at the origin. Translate it to pairwise disjoint balls centered at $3je_n$, $j \geq 1$, obtaining $w_j \in C_c^\infty(X)$. Put $f_j = (-\Delta + 1)w_j$. Translation invariance bounds all $\|f_j\|_2$ by the same constant, and the Dirichlet resolvent sends f_j to w_j . But $\|w_j - w_k\|_2^2 = 2\|w\|_2^2$ for $j \neq k$, so this image has no convergent subsequence. The resolvent is not compact. In the odd Fourier model the multiplier $|\xi|^2$ has no L^2 eigenvector at any fixed value, since each sphere has Lebesgue measure zero. For every $\lambda > 0$, choose infinitely many disjoint small balls in $\{\xi_n > 0, 0 < |\xi|^2 < \lambda\}$. A nonzero smooth Fourier function on each ball, extended oddly into its reflection across $\xi_n = 0$, gives infinitely many mutually orthogonal vectors in the spectral subspace. Its dimension is therefore infinite, and the global finite-rank trace used in (DSP32) is unavailable. Nevertheless the domain contains all smooth compactly supported zero-trace tests, so Section 10 gives a smooth local sharp-projector kernel up to the plane and the same compact-set derivative estimate.

Exercise 4. A sharp cutoff is not a smooth multiplier. Explain why bounded spectral powers apply to E_λ , although the function $1_{(-\infty, \lambda]}$ is discontinuous, and why a real-axis uniform-resolvent argument would fail even in the bounded Dirichlet case.

Solution. The lower bound a restricts the nonzero spectral measure of $E_\lambda f$ to the bounded interval $[a, \lambda]$. The moment integral in (DSP1) is then at most $\max_{a \leq t \leq \lambda} |t|^{2q} \|f\|^2$, which proves membership in every power domain without any continuity of the cutoff. Conversely, at an eigenvalue λ_j , $(A - z)^{-1}\phi_j = (\lambda_j - z)^{-1}\phi_j$. For $z = \lambda_j + i\eta$, its norm on this unit vector is $|\eta|^{-1}$, diverging as $\eta \rightarrow 0$. Lower-bounded self-adjointness cannot supply a uniform bound across such poles. The construction (DSP42)–(DSP44) avoids that assertion and retains the specified closed endpoint.

Further questions and references

The weighted form, explicit normal sign, one-dimensional integrated-square identity, parameter dilation, and local-domain transposition above state their conventions in full.

Three research routes begin with proved objects in this lesson: derive a leading local spectral asymptotic from the actual wave or elliptic parametrix while retaining the density; determine a remainder uniformly as the observation point approaches the boundary; and replace the scalar Dirichlet test-domain condition by the exact trace pairing and complementing-condition hypotheses for a general elliptic boundary

system. Estimate (DSP34) supplies growth control for these tasks, and by itself supplies neither their leading coefficients nor their general-system proofs.

Editorial supplement: the complete boundary receivers and local trace consequences

This supplement supplies the integer extension, boundary and interior regularity, weak coordinate changes and test-density arguments invoked above. Its proofs are part of this lesson. The linked boundary and local-coefficient lessons identify the corresponding programme objects; their ongoing revisions are not premises of these arguments. The original expression (DSP2), density, forms, shifts, recursive domains, projectors, norms, formula labels and examples above remain unchanged. The full spectral-measure construction used for the arbitrary extension is Spectral measures with the original operator domain retained, Sections 1–4, especially (LB3)–(LB19). The exact Fourier maps and their full inverse factors are proved in Fourier transforms, finite spectra and convex separation, Sections 1–2 and 7–8. This supplement also proves the localized trace that survives on an unbounded domain and two sharp endpoint obstructions.

E1. Integer weak-derivative spaces, density and reflection

Fix an integer $k \geq 0$. First let v have every weak derivative through order k in $L^2(\mathbb{R}_+^n)$. There is a sequence smooth on a neighborhood of the closed half-space converging to v in the actual derivative-sum norm (DSP6). Here is a construction that does not assume an extension theorem or a trace theorem. Multiply first by $\theta(x/R)$, where θ is smooth compactly supported and equals one near zero. In every derivative of the difference, the term with no derivative on the cutoff converges by the L^2 tail of that derivative of v ; the remaining terms contain a factor R^{-j} , $j \geq 1$, and an L^2 derivative of v . They tend to zero. Thus it suffices to approximate a function of compact support.

For $\varepsilon > 0$, put $v_\varepsilon(z, t) = v(z, t + \varepsilon)$ on $t > -\varepsilon$. On $t \geq 0$, each weak derivative is the corresponding translate of the derivative of v . Zero-extend each original derivative only as an L^2 function for this comparison. Translation is continuous in L^2 : its Fourier multiplier $e^{i\varepsilon\xi_n}$ tends to one, is bounded in absolute value, and Plancherel and dominated convergence give the assertion. Consequently $v_\varepsilon \rightarrow v$ in the derivative-sum norm on the positive half-space. Convolve v_ε , extended by zero below $-\varepsilon$, with a compact smooth mollifier of radius $\delta < \varepsilon/2$. On a neighborhood of $t \geq 0$, convolution differentiates against the translated weak derivatives, because the convolution never meets the interface $-\varepsilon$. The convolution is smooth there. For fixed ε , Fourier dominated convergence for each of the finitely many derivatives permits a choice of δ making the derivative-sum error less than ε . Compact support is retained in a fixed slightly larger neighborhood. This proves the asserted density.

For $k \geq 1$, the Vandermonde matrix with entries $(-j)^\ell$, $1 \leq j \leq k$, $0 \leq \ell < k$, is invertible: a polynomial of degree less than k vanishing at its k distinct nodes is zero, successively dividing by each linear factor. Thus the coefficients in (DSP24) exist uniquely. They also have the exact closed formula

$$c_j = \prod_{\substack{1 \leq l \leq k \\ l \neq j}} \frac{1+l}{l-j} = \frac{(-1)^{j-1}(k+1)!}{(j+1)(j-1)!(k-j)!}, \quad 1 \leq j \leq k. \quad (\text{DSP-ECJ})$$

Indeed the Lagrange polynomial $\prod_{l \neq j} (t+l)/(l-j)$ has value one at $-j$ and zero at the other nodes. Subtracting its interpolation sum from any degree-less-than- k polynomial gives a polynomial with those k zeros, hence zero. Evaluate this identity at $t = 1$ for the monomials t^ℓ to obtain exactly the original equations. Counting the $j-1$ negative denominator factors and keeping the omitted $j+1$ numerator factor gives the factorial formula. This evaluates the original coefficients and therefore the full original sums S_{k, β_n} , without changing them.

Apply that reflection to the smooth approximants just constructed. The matching derivatives through order $k - 1$ remove every interface delta, by integration by parts against a test function on each side. The calculation (DSP25) bounds every derivative separately, including its exact dilation factor $j^{\beta_n - 1/2}$. The reflected approximants are Cauchy in the whole-space weak-derivative norm and hence in (RT1), by (RT3). Their limit restricts to v ; their piecewise L^2 limits are exactly (DSP24). It follows that (DSP24)–(DSP25) hold for every such weak-derivative input, without a boundary condition. For $k = 0$, the zero extension preserves the L^2 norm.

On a smooth bounded domain use a finite partition on a neighborhood of its compact closure, with each support compactly inside an interior or flattened boundary chart. Such a partition is constructed as follows. Choose finitely many smaller coordinate balls covering the compact closure, and nonnegative compact smooth bumps σ_a supported in the corresponding larger charts and positive on those smaller balls. Their sum σ is positive on a neighborhood of the closure. Choose a compact smooth θ , equal to one on a smaller neighborhood of the closure, with support in $\{\sigma > 0\}$, and put $\zeta_a = \theta\sigma_a/\sigma$ there and zero outside. The denominator is bounded below on the support of θ ; these are smooth, retain each chart support, and sum exactly to one near the closure. The bumps come from the explicit smooth compact cutoff construction of the Fourier and scalar foundations. Restriction gives a nonnegative surface partition; for the prescribed surface partition in (RT21), the collar reconstruction in Trace proof 6 retains those specified boundary values instead.

Smooth pullback acts boundedly on the weak-derivative norm: the ordinary product and chain rules give (RT19) term by term; to justify them for weak derivatives, mollify on compact subsets of the chart interior, pass the chain-rule identities against interior tests, and use the Jacobian bounds to pass each L^2 term. Every coefficient of the resulting finite sum is bounded on the fixed chart support. That distributional identity and its L^2 bounds hold throughout the open positive chart, including all points arbitrarily close to its wall. Apply the preceding flat density and reflection to each localized piece; multiply its reflection by a larger coordinate cutoff equal to one on the original piece, pull back, and sum. This constructs the extension used after (DSP27) and proves its H^k and L^2 bounds with all the constants of (DSP25), (DSP27a), the larger cutoff, and both chart Jacobians. Its restriction is exactly the original function because the partition sums to one.

For the reverse comparison, restriction of a whole-space H^k function has the specified weak derivatives, and (RT3) bounds their derivative-sum norm. Taking the quotient infimum gives that bound for (RT2). The extension just constructed gives the converse bound. Thus the weak-derivative space and the restriction space agree at every nonnegative integer order, with their stated different norms. Applying the extension and then whole-space smooth approximation proves density by restrictions smooth on a neighborhood of the closure. There is no circular use of (RT4) in this construction. Parts 2–6 of the real-order trace proof can therefore use this integer comparison; (RT15)–(RT20) supply all noninteger composition and atlas comparisons, with their complete constants and edge terms.

E2. A complete local weak H^1 -to- H^2 proof

Work first in an upper half-ball, and use nested patches whose closures stay away from its curved edge. In ordinary derivatives let

$$-\partial_j(a^{jk}\partial_k u) + d^k\partial_k u + eu = f, \quad u \in H^1, \quad \gamma_+ u = 0, \quad a^{jk} = a^{kj} \in W^{1,\infty}, \quad a^{jk}\xi_j\xi_k \geq \vartheta|\xi|^2, \quad d, e \in L^\infty, \quad f \in L^2. \quad (\text{DSP-E1})$$

Here $\vartheta > 0$, the principal coefficients are real, the lower coefficients and the function may be complex, and all bounds are on the larger fixed patch. Put $F = f - d^k\partial_k u - eu$. Then $\|F\|_2 \leq \|f\|_2 + (\sum_k \|d^k\|_\infty + \|e\|_\infty)\|u\|_{H^1}$. The distributional divergence equation gives $\int a^{jk}\partial_k u \overline{\partial_j v} dx = \int F \overline{v} dx$ for compactly supported zero-trace H^1 tests: (RT29) and inward approximation in Trace proof 7, whose integer

comparison was proved in E1, approximate these tests by compact interior smooth ones. Thus these are legitimate weak tests, with no assumed second derivative of u .

Take a tangential direction e_i , $i < n$, and $\delta_h w(x) = (w(x + he_i) - w(x))/h$. Choose a real smooth $0 \leq \eta \leq 1$, equal to one on the smaller patch and supported in the larger one, and take $|h|$ smaller than its fixed distance from the curved edge. The test $-\delta_{-h}(\eta^2 \delta_h u)$ has zero trace and the required support. Translation preserves the zero flat trace, by (RT8) in the tangential variables and density. For a fixed $h \neq 0$ the test belongs to H^1 , even before its uniform bound is proved. Changing the integration variable gives the exact discrete integration identity

$$\begin{aligned} \int a^{jk}(x + he_i) \partial_k \delta_h u \overline{\partial_j(\eta^2 \delta_h u)} dx &= \int F \overline{-\delta_{-h}(\eta^2 \delta_h u)} dx \\ &\quad - \int (\delta_h a^{jk})(x) \partial_k u(x) \overline{\partial_j(\eta^2 \delta_h u)} dx. \end{aligned} \quad (\text{DSP-E2})$$

All integrals are over the flat patch, or its zero extension where the test vanishes. No normal translation crosses the wall.

For any tangentially supported H^1 function w , $\delta_h w(x) = \int_0^1 \partial_i w(x + she_i) ds$ in L^2 . This follows first for smooth functions from the fundamental theorem of calculus, and then from E1 density. Minkowski or Cauchy–Schwarz in s gives $\|\delta_h w\|_2 \leq \|\partial_i w\|_2$, with the derivative on the translated larger patch. Also $\|\delta_h a^{jk}\|_\infty \leq \|\partial_i a^{jk}\|_\infty$; the same integral identity for Lipschitz coefficients holds on almost every line, proved by one-dimensional smooth convolution and its weak derivative limit. Therefore

$$\begin{aligned} \|\delta_h u\|_2 &\leq \|\partial_i u\|_{2, \text{larger}}, \\ \|\nabla(\eta^2 \delta_h u)\|_2 &\leq \|\eta \nabla \delta_h u\|_2 + 2\|\nabla \eta\|_\infty \|\delta_h u\|_2, \\ \|\delta_{-h}(\eta^2 \delta_h u)\|_2 &\leq \|\partial_i(\eta^2 \delta_h u)\|_2. \end{aligned} \quad (\text{DSP-E3})$$

The principal term of the real part of (DSP-E2) is at least $\vartheta \|\eta \nabla \delta_h u\|_2^2$. Its cutoff terms are bounded by $2 \sum_{jk} \|a^{jk}\|_\infty \|\nabla \eta\|_\infty \|\eta \nabla \delta_h u\|_2 \|\delta_h u\|_2$. Using (DSP-E3), the right side is bounded by a fixed constant times $(\|F\|_2 + \|u\|_{H^1})(\|\eta \nabla \delta_h u\|_2 + \|u\|_{H^1})$. Apply Young's inequality with coefficient $\vartheta/2$ to the terms linear in the first parenthesized gradient norm, and move that half to the left. This gives

$$\|\eta \nabla \delta_h u\|_2 \leq C(\|f\|_{2, \text{larger}} + \|u\|_{H^1(\text{larger})}), \quad (\text{DSP-E4})$$

where C depends only on ϑ^{-1} , the principal $W^{1,\infty}$ bounds, the lower L^∞ bounds, the cutoff and the nested patches. Density and the weak trace have supplied the domain of every test; no differentiability being established was assumed.

For each j , (DSP-E4) bounds the tangential difference quotients of $\partial_j u$ on the smaller patch. To obtain its weak derivative without invoking a compactness theorem, test against $\psi \in C_c^\infty$ there. Discrete integration and the convergence of the smooth test differences give

$$\left| \int \partial_j u \overline{\partial_i \psi} dx \right| = \lim_{h \rightarrow 0} \left| \int \delta_h \partial_j u \overline{\psi} dx \right| \leq C(\|f\|_2 + \|u\|_{H^1}) \|\psi\|_2. \quad (\text{DSP-E5})$$

The Hilbert representation proved before (DSP5), and density of tests in L^2 , supply an L^2 distributional derivative $\partial_i \partial_j u$ with exactly this bound. Commuting weak derivatives gives every second derivative with at least one tangential index.

It remains to recover the pure normal derivative. Put $p = a^{nn} \partial_n u$. In distributions the equation gives

$$\partial_n p = -f + d^k \partial_k u + eu - \sum_{(j,k) \neq (n,n)} (a^{jk} \partial_j \partial_k u + (\partial_j a^{jk}) \partial_k u). \quad (\text{DSP-E6})$$

Every term on the right is now L^2 . The weak product rule and $a^{nn} \geq \vartheta$ give

$$\partial_n^2 u = (a^{nn})^{-1} \partial_n p - (a^{nn})^{-2} (\partial_n a^{nn}) p \in L^2. \quad (\text{DSP-E7})$$

To justify that rule, multiply an H^1 function by a Lipschitz coefficient, approximate the coefficient by bounded smooth convolutions, and integrate against compact tests. The coefficients converge uniformly and their derivatives converge weakly against L^1 test products, giving the displayed product derivative. The same proof works when only the normal derivative is known, using one-dimensional convolution on normal lines and Fubini. Apply it to p and $1/a^{nn}$, both of which have the required normal derivative and bounded multiplier. Thus multiplying an undefined second derivative by a rough coefficient was not a step of the proof. In dimension one, (DSP-E6)–(DSP-E7) give the entire regularity proof directly.

We have proved on the smaller half-patch

$$\|u\|_{H^2(\text{smaller})} \leq C(\|f\|_{2,\text{larger}} + \|u\|_{H^1(\text{larger})}). \quad (\text{DSP-E8})$$

On an interior patch the same difference-quotient proof uses all directions, because every cutoff test is supported in the interior. Equation (DSP-E5) then gives all second derivatives without (DSP-E6). This proves the exact interior counterpart locally, under the same Lipschitz principal and bounded lower bounds. Neither proof requires symmetry of the lower terms. In particular it applies after either smooth or $C^{1,1}$ flattening of the original weighted symmetric expression.

E3. Weak charts, zero-trace test density and smooth bootstrap

For the $C^{1,1}$ graph in (DSP15b), both F and F^{-1} are locally bi-Lipschitz, have bounded first derivatives and bounded weak second derivatives. Their determinant is exactly one. The first weak chain rule for H^1 inputs follows by interior smooth approximation and change of variables. For a Lipschitz chart one can first convolve its graph function: the resulting smooth charts have the same determinant and uniformly bounded derivatives, converge with their first derivatives uniformly on compact patches, and have a common inverse Lipschitz bound. Composition converges in L^2 , first for continuous compact smooth inputs, then for general L^2 inputs by density and the uniform Jacobian bound. Passing the first chain-rule identity against tests gives the rule for the actual chart. For an H^2 input apply this first rule also to each first derivative, and use the weak Lipschitz product rule from E2 on each bounded first derivative of the chart. This gives exactly (DSP15d), with both displayed sums. It proves bounded H^2 pullback in both directions, with constants containing the first and second weak derivatives and both Jacobian bounds. The half-space H^2 approximation and extension required here were established in E1.

Changing variables in the weak divergence pairing gives (DSP15c), with $M = (DF)^{-1}$, $M(g \circ F)M^T$, and $\tilde{r} = (r \circ F)J_F$ in their displayed order. This matrix is real symmetric and Lipschitz, and its quadratic form is at least $\theta|M^T\xi|^2$. Its ellipticity lower bound is therefore $\theta\|M^{-T}\|_\infty^{-2}$. The original lower coefficients remain bounded after their indicated pullback, and expansion of the divergence retains the bounded first derivatives of this matrix. The lower and upper transformed density bounds are those of the actual product $(r \circ F)J_F$; none is deleted. The trace-kernel proof in the paragraph after (DSP15c) uses only the now proved H^1 chain rule and (RT18) with $m = 0, \sigma = 1/2$, so it remains valid. Applying (DSP-E8), pulling back by (DSP15d), and using a finite cover proves (DSP15a) exactly. The inverse and compactness arguments in Sections 1 and 5 require only the form bounds and zero extension of the defining H_0^1 approximants, and consequently hold on this same $C^{1,1}$ domain.

For clarity the graph example in the original figure can be calculated entirely here. In two coordinates, for $h(z) = z|z|$,

$$h' = 2|z|, \quad h'' = 2 \operatorname{sgn} z \text{ almost everywhere,} \quad M = \begin{pmatrix} 1 & 0 \\ -2|z| & 1 \end{pmatrix}, \quad MM^T = \begin{pmatrix} 1 & -2|z| \\ -2|z| & 1 + 4z^2 \end{pmatrix}. \quad (\text{DSP-E9})$$

With the original $P = -\Delta$ and $r = 1$, its exact expression is

$$(Pu) \circ F = -\partial_z^2 v + 4|z|\partial_z \partial_t v - (1 + 4z^2)\partial_t^2 v + 2\operatorname{sgn} z \partial_t v = -\partial_a((MM^T)^{ab}\partial_b v). \quad (\text{DSP-E10})$$

The weak second derivative of h is the displayed bounded function; it has no delta because h' is continuous. The determinant of the matrix is one, while its entries and the weak drift remain exactly as displayed. This proves the figure's local example without assuming the pending boundary lesson as a provider.

Next let $v \in H^2(\mathbb{R}_+^n)$ have zero trace, localized to a compact half-patch. Tangential trace commutation follows by the smooth density of E1, the order-two trace estimate (RT9), and the continuous derivative $H^{3/2}(\mathbb{R}^{n-1}) \rightarrow H^{1/2}(\mathbb{R}^{n-1})$, whose norm bound follows directly from its Fourier multiplier. Hence $\gamma_+ \partial_i v = \partial_i \gamma_+ v = 0$ for $i < n$. Define its odd extension

$$Ov(z, t) = \begin{cases} v(z, t), & t > 0, \\ -v(z, -t), & t < 0. \end{cases} \quad (\text{DSP-E11})$$

Integration by parts on both sides gives no first-derivative interface delta because both traces of v are zero. The normal first derivative extends evenly and has matching traces; each tangential first derivative extends oddly and has zero trace on both sides. Thus all second-derivative interface deltas also vanish. Each derivative through order two has whole-space squared norm exactly twice its positive-half-space norm. Formula (DSP-E11) is therefore an H^2 extension. Convolution with a smooth mollifier even in t preserves oddness; a compact cutoff even in t , equal to one near the original support, preserves it too. Their restrictions are smooth with zero trace and converge in H^2 . Pull back and sum over a smooth boundary partition, treating interior pieces by ordinary convolution. This proves

$$\overline{\{v \in C^\infty(\bar{\Omega}) : \gamma v = 0\}}^{H^2} = H^2(\Omega) \cap H_0^1(\Omega) \quad (\text{DSP-E12})$$

on every bounded smooth auxiliary domain used in Section 10. The inclusion from left to right follows by trace continuity. The reverse inclusion is the construction just given. On a noncompact half-space the same proof uses expanding compact cutoffs first. Consequently the odd Fourier realization in Exercise 3 has exactly the asserted $H^2 \cap H_0^1$ domain, rather than merely an agreement on interior tests.

For smooth coefficients and charts, the bootstrap in (DSP10)–(DSP13) now has its actual first regularity input (DSP-E8). At its induction level k , a purely tangential derivative $D_z^\beta u$, $|\beta| = k$, is H^1 by the already proved H^{k+1} bound. Trace commutation is justified as above at order $k+1$: (RT9) and the Fourier multiplier D_z^β commute on smooth approximants and pass in $H^{1/2}$. Its trace is zero. After a fixed smaller cutoff, its equation has L^2 right side by the full sum (DSP12), with the cutoff contributions (DSP12c). Apply (DSP-E8) to it. This bounds all derivatives of total order $k+2$ with at most two normal factors, including the case of zero tangential factors at the base level. Then the exact ordinary or D -normal equation (DSP11) and its full Leibniz sum (DSP12a) give the successive estimates (DSP12b) for three through $k+2$ normal factors. Derivatives on the reciprocal of g^{nn} are retained by the ordinary quotient and product rules; their bounds depend on the original positive lower bound and the coefficient derivatives through the displayed order. In dimension one the same normal recurrence alone gives every order. Interior differentiation uses all directions and (DSP-E8), without a trace requirement. The finite nested-patch construction and the original form bound therefore prove (DSP8)–(DSP9) at every stated integer order, with the sufficient coefficient and chart bounds in Exercise 2.

For the first domain, the forward implication is (DSP-E8) plus the form estimate, and the reverse is expansion of the original expression on $H^2 \cap H_0^1$. For its powers no boundary condition is inferred merely from a scalar moment. Instead the recursion in Section 4 successively applies that first-domain

equality to each actual $P_D^j u$, proving (DSP14)–(DSP15) in both directions. The simultaneous binomial identities on each recursive domain prove $D((P_D + \mu)^m) = D(P_D^m)$, retaining the lower powers. This connects the full boundary graph to the spectral moments in (DSP16b)–(DSP16c).

E4. The arbitrary extension's domain is obtained before its boundary trace

Keep the full original expression and density on an allowed smooth patch. Set $a^k = b^k + r^{-1}(\partial_j r)g^{jk}$. Writing the expression without changing it gives $P = -r^{-1}\partial_j(rg^{jk}\partial_k) + a^k\partial_k + c$. Its weighted formal adjoint on compact interior tests is

$$P^\dagger = -r^{-1}\partial_j(rg^{jk}\partial_k) - \overline{a^k}\partial_k + \overline{c} - r^{-1}\partial_k(r\overline{a^k}). \quad (\text{DSP-E13})$$

Integration by parts retains both density derivative terms. Equality of the test operators gives $a^k = -\overline{a^k}$ and $c - \overline{c} = r^{-1}\partial_k(r\overline{a^k})$: to identify coefficients, take a test equal to a prescribed affine polynomial near any chosen interior point, and then a test constant there, in the resulting first-order difference. Smoothness gives equality at the point, hence everywhere. Changing variables in this actual pairing yields precisely (DSP37a), with the transformed Jacobian already in $\tilde{r}$; no unweighted adjoint is substituted.

For an arbitrary $u \in D(A)$, only the given inclusion (DSP33) is used. The auxiliary smooth domain in Section 10 has the exact flattened defining function displayed there, is contained in the physical one-sided domain, and agrees with its boundary where the real cutoff is supported. All coefficient, density and chart bounds are finite on its compact closure. Its form is Hermitian on H_0^1 , by the compact-test symmetry and density, and a shift σ makes it coercive as in Section 1. The functional (DSP36) is bounded in the H^1 norm, because the commutator is exactly the order-one expression before (DSP12c). Hilbert representation produces w and the bound (DSP37). For every smooth zero-trace test, the supported function χv belongs to the prescribed domain class, so the adjoint identity for the original $A = A^*$ proves (DSP38). Integration by parts with compact approximants to w proves the corresponding form identity. Density (DSP-E12) extends both identities to every $H^2 \cap H_0^1$ test. The full original operator is continuous from that space to the weighted L^2 space, with exactly the coefficient sum (DSP37b).

For every $f \in L_r^2(\Omega)$, the same coercive form solves its Dirichlet equation in H_0^1 . Applying (DSP-E8) on a finite boundary/interior cover places that solution in H^2 , so $P_{D,\Omega} + \sigma$ is onto from $H^2 \cap H_0^1$. Subtract the two test identities and use this surjectivity: their pairing with every f is zero. Hence $\chi u = w$ in the original weighted L^2 space. This proves local H^1 , its bound, and zero trace before any boundary value of u was assumed. On the smaller patch $Pu = Au \in L^2$; (DSP-E8) proves (DSP35). Its higher receivers follow from E3 on nested patches applied to $Pu = Au$ and the actual recursion $Au \in D(A^{q-1})$. This proves (DSP39), including each cutoff commutator and every power on its original domain. The argument does not require multiplication by χ to preserve $D(A)$.

The accepted spectral construction cited at the start of this supplement proves, on exactly this weighted Hilbert space, the inverse of $A + \rho$, its lower concentration, the order-reversing map $t \mapsto t^{-1} - \rho$, equality of the original operator graph, and both recursive power conventions. Its (LB17) keeps both closed interval endpoints, including any atom at λ . Therefore (DSP40) is the actual spectral-power estimate, and (DSP19) applied to localized functions gives (DSP41) with no additional exponent. Real coordinate Taylor remainders (DSP43a)–(DSP43e) then prove the Hilbert derivatives of the exact Riesz vectors, including the continuous one-sided jets on a curved wall. With the linear-first inner product, (DSP44) is $(h_y, h_x)_r$, and every mixed derivative is in its displayed order. This supplies a complete receiver for (DSP34) and (DSP45), for every finite real cutoff, on every allowed compact set. If $\lambda < a$, its projection and all these vectors are exactly zero. If $\lambda \geq a$ is less than one, $s = \max(1, \lambda + \rho)$ in the same proved parameter estimate gives the asserted finite smooth norms. None of these steps replaces a sharp cutoff by a smooth approximation.

The further Hilbert-valued extension check after (DSP43e) also follows from the same scalar proofs. For a compactly localized Hilbert-smooth map, expand it in an orthonormal basis of the separable closed span of its values and derivatives. Such a span is separable because each continuous derivative map on a compact coordinate set has a separable image, by finite nets, and there are only countably many derivative orders. Finite basis projections converge in the L^2 norm of each derivative by Parseval and dominated convergence. Apply (DSP24)–(DSP25) to every scalar component, and sum their squared norms by Tonelli; this gives the same Hilbert-valued reflection bounds and the same matching jets without changing any coefficient. Apply scalar Plancherel to the finite projections; completeness and their exact norm factors construct the Hilbert-valued Fourier transform and its inverse. The pairing with every unit vector and Cauchy–Schwarz with the scalar weight in (DSP27) give its vector-valued derivative bound. Fourier dominated convergence, with the integrable vector norm supplied by that same inequality, gives the continuous derivatives. Thus this additional extension argument uses no unproved vector-valued transform or extra trace assumption.

E5. A localized trace remains available without compact resolvent

Let A , r , E_b and the allowed patches be exactly those of Section 10. Let χ be any complex bounded measurable function whose essential support is contained in a compact union of allowed patches, and let M_χ denote its multiplication on the original space H . This includes each smooth cutoff already used above; this trace proof differentiates neither χ nor its kernel. The map $R = M_\chi E_b$ is Hilbert–Schmidt even when E_b has infinite rank. More precisely,

$$\|M_\chi E_b\|_2^2 = \int_X |\chi(x)|^2 e_A(x, x, b) r(x) dx, \quad M_\chi E_b M_{\bar{\chi}} \in \mathcal{S}_1(H), \quad \text{Tr}(M_\chi E_b M_{\bar{\chi}}) = \|M_\chi E_b\|_2^2. \quad (\text{DSP-E14})$$

For $b < a$ every term is zero; for every other finite b it is finite. If $b \geq \max(1, a)$, (DSP43) gives the bound

$$\text{Tr}(M_\chi E_b M_{\bar{\chi}}) \leq C_{\text{supp } \chi} (b + \rho)^{n/2} \int_X |\chi|^2 r dx. \quad (\text{DSP-E15})$$

All measures and both multiplication factors are retained. The trace is localized in the physical variable; this does not assert a finite global trace.

Here is the complete kernel-to-ideal argument. The space $L^2(X, r dx)$ is separable. Use the fixed compact exhaustion $K_m = \{x \in X : |x| \leq m, \text{dist}(x, \mathbb{R}^n \setminus X) \geq 1/m\}$, interpreting the distance to the empty set as infinity. These sets are increasing and cover X ; the density is bounded above and below on each nonempty K_m . Dominated convergence gives $f 1_{K_m} \rightarrow f$ in the original weighted norm. Its zero extension is unweighted L^2 , so the already proved compact smooth density gives whole-space smooth approximants. On a bounded box their uniform approximation by rational complex values on a finite rational box mesh follows from uniform continuity. Restrict these approximants to K_m ; the weighted error is bounded by its actual $r_{\max, K_m}^{1/2}$ times the unweighted error. The countable family of all such finite rational meshes and values, restricted to the countably many fixed K_m , is consequently dense. Apply Gram–Schmidt to a dense sequence, discarding zero remainders; its span is dense, and the Hilbert expansion proof of (DSP16a) proves Parseval. Write this countable orthonormal basis as (v_j) . Since evaluation on the range has representative $h_x^{(b)}$, Parseval and Tonelli give

$$\sum_j \|M_\chi E_b v_j\|_H^2 = \int_X |\chi|^2 \sum_j |(v_j, h_x^{(b)})_r|^2 r dx = \int_X |\chi|^2 \|h_x^{(b)}\|_H^2 r dx. \quad (\text{DSP-E16})$$

The integrand is bounded on its compact support, and its last norm is $e_A(x, x, b)$, so this is finite. The square-sum is the Hilbert–Schmidt norm as defined and proved basis-independent in (T1)–(T4) of

Traces that survive passage to cohomology. Its factorization theorem (T5)–(T9) says that a product of two such factors is trace class, with norm at most the product of their square-sum norms. Apply it to RR^* ; the trace proof (T10)–(T13) gives $\text{Tr } RR^* = \sum_j \|Rv_j\|^2$. These are complete earlier programme proofs, not external trace theorems. Alternatively, the same identity follows by the convergent positive rank-one expansion $RR^* = \sum_j (\cdot, Rv_j)_r Rv_j$, whose sum of rank-one trace norms is (DSP-E16). This proves every assertion in (DSP-E14)–(DSP-E15).

The bounded multiplier class (DSP45a) has a further exact local trace consequence. For any such m ,

$$\begin{aligned} M_\chi m(A) M_{\bar{\chi}} &\in \mathcal{S}_1(H), \\ \|M_\chi m(A) M_{\bar{\chi}}\|_1 &\leq \|m\|_{F_{A,\infty}} \int_X |\chi|^2 e_A(x, x, b) r \, dx, \\ \text{Tr}(M_\chi m(A) M_{\bar{\chi}}) &= \int_X |\chi(x)|^2 e_m(x, x) r(x) \, dx. \end{aligned} \tag{DSP-E17}$$

Indeed it equals $Rm(A)R^*$, where $R = M_\chi E_b$, because the support condition gives exactly $m(A) = E_b m(A) E_b$. Its two Hilbert–Schmidt factors prove the norm bound. For the diagonal identity expand the trace of a product of Hilbert–Schmidt factors in an orthonormal basis twice. The sum of the absolute values of its matrix products is at most the product of the two Hilbert–Schmidt norms, by Cauchy–Schwarz on the two indices. Interchanging those absolutely convergent sums proves $\text{Tr}(Rm(A)R^*) = \text{Tr}(m(A)R^*R)$. Bounded cyclicity (T15), with R^*R trace class, then gives $\text{Tr}(R^*Rm(A))$. In this last basis expansion, each term is $\int |\chi(x)|^2 (m(A)v_j, h_x^{(b)})_r (h_x^{(b)}, v_j)_r r(x) \, dx$. At each x , the sum is absolutely bounded by $\|m\|_{F_{A,\infty}} \|h_x^{(b)}\|^2$, by Cauchy–Schwarz and Parseval applied also to $m(A)^* h_x^{(b)}$. This integrable bound permits exchange of sum and integral. The sum is exactly $(m(A)h_x^{(b)}, h_x^{(b)})_r = e_m(x, x)$. This proves (DSP-E17) for complex, discontinuous multipliers, with their full essential spectral bound. Thus the lesson’s infinite-rank example still has a finite trace after this exact physical localization.

E6. Explicit sharp half-space kernel, density and boundary comparison

In Exercise 3, let $X = \mathbb{R}_+^n$, $P = -\Delta$, and first keep $r = 1$. Odd extension divided by $\sqrt{2}$ is the unitary map specified there. Its image is the closed odd subspace. Fourier multiplication by $|\xi|^2$ is self-adjoint on the exact domain $\{U : |\xi|^2 \widehat{U} \in L^2\}$, by testing coordinate multiplication and its conjugate; (RT1)–(RT3) identify that domain with $H^2(\mathbb{R}^n)$. It preserves oddness, as do its spectral projections. E3 proves that its transported domain is exactly $H^2(X) \cap H_0^1(X)$. The inverse shifted multiplier $(|\xi|^2 + 1)^{-1}$ gives the original Dirichlet inverse there, so no different realization is introduced.

Put $J = \text{diag}(1, \dots, 1, -1)$, $y^* = Jy$. Folding the whole-space Fourier projection integral across the wall, with both factors of the unitary extension and restriction retained, gives

$$e_D(x, y, \lambda) = (2\pi)^{-n} \int_{|\xi|^2 \leq \lambda} (e^{i(x-y)\cdot\xi} - e^{i(x-Jy)\cdot\xi}) \, d\xi \quad (\lambda \geq 0). \tag{DSP-E18}$$

For $\lambda < 0$ it is zero. The set at $\lambda = 0$ has measure zero, so the closed endpoint is still retained and gives zero. To justify the kernel identity initially take compact smooth f in X . Its odd extension is $L^1 \cap L^2$; Fourier inversion of the bounded-frequency projection is legitimate by Cauchy–Schwarz on the finite ball. Fubini gives (DSP-E18) against dy . The kernel row is in $L^2(X)$, as follows from whole-space Plancherel for that same finite-ball multiplier and reflection. Approximation then proves the identity for every original L^2 datum in its pairing sense.

Differentiating the finite integral gives the full signed formula

$$D_x^\alpha D_y^\beta e_D = (2\pi)^{-n} \int_{|\xi|^2 \leq \lambda} \xi^\alpha [(-\xi)^\beta e^{i(x-y)\cdot\xi} - (-J\xi)^\beta e^{i(x-Jy)\cdot\xi}] \, d\xi. \tag{DSP-E19}$$

Both terms remain. Consequently, for $\lambda > 0$,

$$|D_x^\alpha D_y^\beta e_D| \leq 2(2\pi)^{-n} \lambda^{(n+|\alpha|+|\beta|)/2} \int_{|\eta| \leq 1} |\eta^\alpha \eta^\beta| d\eta. \quad (\text{DSP-E20})$$

This proves the exponent of (DSP34) on this exact continuous-spectrum model for every positive λ . Reflection in ξ_n shows that (DSP-E18) vanishes when either point lies on the wall. Its diagonal is

$$e_D(x, x, \lambda) = (2\pi)^{-n} \int_{|\xi| \leq \sqrt{\lambda}} (1 - \cos(2x_n \xi_n)) d\xi \geq 0. \quad (\text{DSP-E21})$$

It need not be zero in the interior. The oscillatory integral tends to zero as $x_n \rightarrow \infty$: approximate the ball indicator in L^1 by compact smooth functions; for each such function integration by parts in ξ_n bounds its integral by $(2x_n)^{-1} \|\partial_n \varphi\|_1$, while its L^1 approximation error bounds the remaining integral. Thus the diagonal tends to the positive constant $(2\pi)^{-n} \omega_n \lambda^{n/2}$. Its integral over the whole half-space is infinite for every $\lambda > 0$. Formula (DSP-E14) gives its exact finite localized trace, including the boundary loss, without attributing a global finite trace to it.

The original density can also be a constant $r_0 > 0$ without changing $-\Delta$ or its Dirichlet domain. The isometry $f \mapsto \sqrt{r_0} f$ to the unweighted space intertwines the same operator and its closed-endpoint projectors; putting it on both sides of the integral representation gives

$$e_{D,r_0}(x, y, \lambda) = r_0^{-1} e_D(x, y, \lambda), \quad E_\lambda f(x) = \int_X e_{D,r_0}(x, y, \lambda) f(y) r_0 dy. \quad (\text{DSP-E22})$$

This is an explicit comparison map; the original weighted object and the r_0 in its measure remain displayed. In dimension one write $a = \sqrt{\lambda}$. Direct integration of (DSP-E18) gives

$$\begin{aligned} e_{D,r_0}(x, y, \lambda) &= \frac{1}{\pi r_0} \left[\frac{\sin(a(x-y))}{x-y} - \frac{\sin(a(x+y))}{x+y} \right], \\ e_{D,r_0}(x, x, \lambda) &= \frac{1}{\pi r_0} \left[a - \frac{\sin(2ax)}{2x} \right]. \end{aligned} \quad (\text{DSP-E23})$$

Every zero denominator in these continuous formulas means its actual limit $\sin(at)/t \rightarrow a$; at the wall the two contributions cancel exactly. For the bounded measurable localization $\chi = 1_{(0,L)}$, $L \geq 0$, (DSP-E14) and the substitution $t = 2ax$ give

$$\text{Tr}(M_{1_{(0,L)}} E_\lambda M_{1_{(0,L)}}) = \int_0^L e_{D,r_0}(x, x, \lambda) r_0 dx = \frac{aL}{\pi} - \frac{1}{2\pi} \text{Si}(2aL), \quad \text{Si}(z) = \int_0^z \frac{\sin t}{t} dt. \quad (\text{DSP-E23a})$$

The integrand at zero has its continuous value one. For $a = 0$ or $L = 0$ the trace is zero; these cases also follow directly before the substitution. Both density factors and the entire boundary term remain explicit. This model keeps the reflected boundary term, the Fourier factor, the density and the parameter, and verifies Exercise 3's surviving kernel assertions directly.

E7. The two strict Sobolev endpoints are necessary

The real-order trace threshold cannot be weakened to $s = 1/2$, even if the target is merely the boundary distributions with their weak topology. Choose smooth $0 \leq \psi_R \leq 1$ supported in $(1/2, 2R)$, equal to one on $[1, R]$, with $R \geq 4$. Put

$$I_R = \int_{\mathbb{R}} \frac{\psi_R(\tau)}{\sqrt{1+\tau^2}} d\tau, \quad \hat{g}_R(\tau) = \frac{2\pi}{I_R} \frac{\psi_R(\tau)}{\sqrt{1+\tau^2}}. \quad (\text{DSP-E24})$$

Finite physical traces on infinite spectral subspaces

Original operator: $-d^2/dx^2$ on $x > 0$; Dirichlet boundary; density $r = 2$; $a = \sqrt{\lambda}$

$$e(x, y, \lambda) = [\sin(a(x-y))/(x-y) - \sin(a(x+y))/(x+y)] / (2\pi)$$

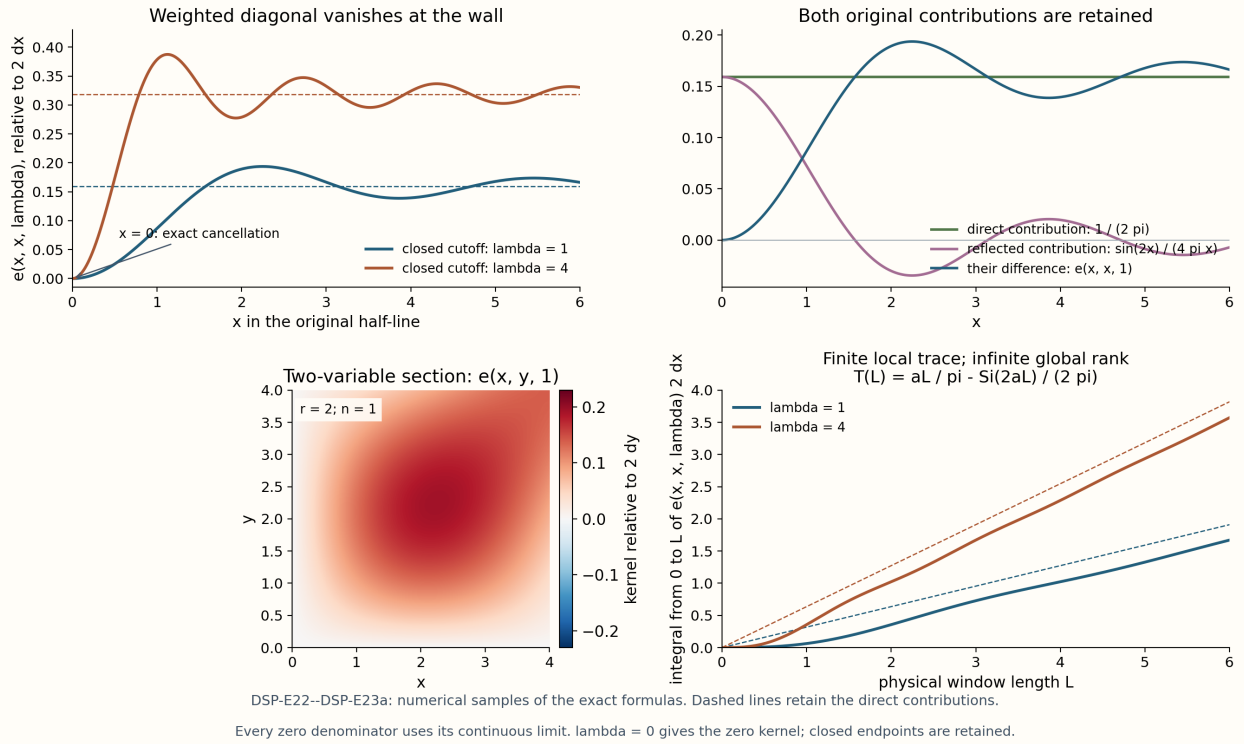


Figure 3: Exact one-dimensional continuous-spectrum Dirichlet projectors on $x > 0$, with original density $r_0 = 2$. The diagonal curves and the kernel section use (DSP-E22)–(DSP-E23), retain the reflected wall term, and include the cutoff endpoints. The physical localized trace uses $e_{D,2}(x, x, \lambda) 2\pi dx$, as proved in (DSP-E14). The drawn kernel is a two-variable section of this one-dimensional model.

These are smooth compact Fourier functions, hence g_R is Schwartz. Fourier inversion gives $g_R(0) = 1$ exactly. The bounds $c \log R \leq I_R \leq C \log R$ follow by integration of τ^{-1} on $[1, R]$, with the two fixed-ratio edge intervals retained in the upper bound. Also

$$\|g_R\|_{H^{1/2}(\mathbb{R})}^2 = \frac{2\pi}{I_R^2} \int \frac{\psi_R(\tau)^2}{\sqrt{1+\tau^2}} d\tau \leq \frac{2\pi}{I_R} \rightarrow 0. \quad (\text{DSP-E25})$$

For $n \geq 2$, choose a fixed nonzero tangential Schwartz function f with compact Fourier support. Then $U_R(z, t) = f(z)g_R(t)$ has $H^{1/2}(\mathbb{R}^n)$ norm tending to zero: on that support $(1 + |\zeta|^2 + \tau^2)^{1/2} \leq C_f(1 + \tau^2)^{1/2}$, and the full Fourier norm and its $(2\pi)^{-n}$ factor separate by Fubini. Yet its classical trace is always f . For $n = 1$, take the single boundary point and use g_R alone. Restriction cannot increase the quotient norm. Multiplication by fixed smooth chart cutoffs is bounded at order $1/2$ by (RT18)–(RT20); choosing the normal cutoff equal to one at zero retains a fixed nonzero localized trace. Pull back in a smooth boundary chart by the proved coordinate comparison. Hence on every smooth boundary patch there are smooth inputs tending to zero in $H^{1/2}(X)$, with the same nonzero smooth trace. No continuous extension of classical restriction to boundary distributions is possible. Since the H^s norm is bounded by the $H^{1/2}$ norm for $0 \leq s \leq 1/2$, the same counterexample applies throughout that range. Thus (RT4)'s strict inequality is sharp for this original restriction-space class.

The strict point-evaluation inequality behind (DSP19)–(DSP20) is also necessary. Fix a multiindex α , put $s_* = |\alpha| + n/2$, and choose a nonnegative smooth angular cutoff κ supported in a small cone where $|\xi^\alpha| \geq c|\xi|^{|\alpha|}$, and nonzero on a smaller cone. Such a cone is obtained around a unit vector with every coordinate used in α nonzero. Let θ_R be a smooth radial cutoff, supported in $1/2 < |\xi| < 2R$, equal to one for $1 \leq |\xi| \leq R$. Define

$$\begin{aligned} J_R &= (2\pi)^{-n} \int \xi^{2\alpha} (1 + |\xi|^2)^{-s_*} \kappa(\xi/|\xi|) \theta_R(\xi) d\xi, \\ \widehat{V}_R(\xi) &= J_R^{-1} \xi^\alpha (1 + |\xi|^2)^{-s_*} \kappa(\xi/|\xi|) \theta_R(\xi). \end{aligned} \quad (\text{DSP-E26})$$

Take $0 \leq \kappa, \theta_R \leq 1$. Polar integration with its Jacobian r^{n-1} gives $c \log R \leq J_R \leq C \log R$: the radial power at infinity is exactly $r^{2|\alpha|-2s_*+n-1} = r^{-1}$, and the fixed angular integral is positive. For $n = 1$ use the positive ray, so the zero-dimensional angular integral has measure one. Fourier inversion and the convention $D = -i\partial$ give

$$D^\alpha V_R(0) = 1, \quad \|V_R\|_{H^{s_*}}^2 \leq J_R^{-1} \rightarrow 0. \quad (\text{DSP-E27})$$

The norm bound follows by squaring the Fourier expression and using $\kappa^2 \theta_R^2 \leq \kappa \theta_R$; all factors of $(2\pi)^{-n}$ are the same as in J_R . Each input is Schwartz. Multiply it by a compact smooth cutoff identically one near zero, and translate its support into an interior ball of the given domain or half-space. The derivative at that point remains one, and (RT20) bounds the norm by a constant times the vanishing norm in (DSP-E27). These compact interior smooth inputs disprove a bounded derivative evaluation at the critical order. In particular, whenever the critical order is an integer, substituting the parameter value one disproves (DSP19) or (DSP20) with equality in their strict hypothesis. At noninteger critical orders it proves the corresponding precise obstruction on the full real Sobolev scale. The original noncritical estimates, with their exact parameter powers and boundary contributions, remain valid.

For clarity the exact Fourier coefficient in the Cauchy–Schwarz step (DSP27) can be chosen as $C_n = (2\pi)^{-n}$. Indeed the squared inverse factor is $(2\pi)^{-2n}$; placing one factor $(2\pi)^{-n}$ in the second energy integral leaves this coefficient in front of the first integral. Its scaling is the exact equality

$$\int \frac{|\xi|^{2|\alpha|}}{|\xi|^{2k} + s^k} d\xi = s^{|\alpha|+n/2-k} \int \frac{|\eta|^{2|\alpha|}}{|\eta|^{2k} + 1} d\eta. \quad (\text{DSP-E28})$$

It has the full Jacobian $s^{n/2}$, numerator $s^{|\alpha|}$ and denominator s^k . Its finite integral is the exact constant used before the multinomial and reflection comparisons; every dilation and derivative coefficient in those comparisons remains in (DSP25) and (DSP27a).

E8. Propagation through the original conclusions

E1 supplies every integer weak-derivative, reflection, density and coordinate receiver used in (RT3), (RT5), (DSP24)–(DSP27a) and Exercise 3. The complete real-order proof above then retains its common lift (RT11)–(RT14), every fractional constant (RT15)–(RT18), each atlas and surface Jacobian, and its full one-sided and zero-trace statements. E2–E3 supply (DSP8)–(DSP15d), all smooth-domain powers, the exact $C^{1,1}$ first domain and the original graph example. They supply the auxiliary onto map and test density in E4, hence (DSP35) and all of (DSP39) on the actual arbitrary extension domain. The explicit spectral construction retains (DSP1), (DSP16a)–(DSP16g), (DSP40), both sharp endpoints and every original shift. The parameter proof with its negative boundary square (DSP21), Vandermonde dilation factors, full derivative sums and Fourier scaling proves (DSP19)–(DSP31) exactly; E7 determines the two critical defects without changing those statements. Weighted orthonormality proves (DSP17)–(DSP18), the derivative bounds and the finite-rank counting bound (DSP32) with multiplicities. E4 supplies the arbitrary sharp and bounded-multiplier kernel conclusions (DSP34), (DSP42)–(DSP45c); E5 strengthens their receiving scope by proving the physical localized trace (DSP-E14)–(DSP-E17), even for infinite global rank. E6 verifies the original noncompact-resolvent exercise and gives its exact reflected kernel, density and diagonal. The other three solutions are the displayed finite projection calculation, full divergence commutator/normal equation, and the exact closed-endpoint moment and eigenvalue-pole calculation. Every exercise therefore uses a proved receiver on its stated domain.

The localized trace and the endpoint obstructions are editorial consequences. No leading spectral asymptotic, general elliptic boundary-system theorem or new literature priority claim is asserted. The original further questions remain further calculations beyond the proved growth bound.

Editorial extension: the full space of physical localization weights

E5 shows that a sharp spectral projector can have a finite trace after multiplication by a bounded weight supported on a compact set, even when its global rank is infinite. This supplement determines the full class of measurable physical weights for which the multiplication–projector map is Hilbert–Schmidt. The criterion is the integral of the squared weight against the original weighted spectral diagonal. It also proves the initial multiplication domains, their exact bounded closures, and the trace for two different complex weights.

The final example keeps the original half-line Dirichlet operator and the complete reflected kernel from E6. The weight is singular at the boundary and extends over the entire half-line, yet its localized trace is finite. The reflection term is precisely what makes the boundary integral converge. The original statements (DSP-E14)–(DSP-E23a), including their bounded physical window, remain the objects to which these additional formulas are compared.

The proofs use the following earlier lessons at these exact locations: Spectral measures with the original operator domain retained, Section 4, (LB14)–(LB19), and Section 6.1, (HF1)–(HF5), for the original projection-valued calculus, Hilbert expansions and bounded adjoints; Traces that survive passage to cohomology, Sections 1–3, (T1)–(T15), for Hilbert–Schmidt products, trace norms, rank-one traces and trace continuity; Fourier transforms, finite spectra and convex separation, Section 7, (FL4)–(FL5), for the full Fourier Plancherel factor; Metric and topological foundations, Section 13.6, (OC22), for the complete finite Taylor remainder used at the zero endpoint; and Banach estimates, quotient spaces and compact parameter arguments, Sections 15.1–15.3 and 16.4–16.5, for dominated convergence, the

original completed measure and nonnegative Tonelli. Evaluation representatives and their local bounds are those proved in this lesson at (DSP42)–(DSP45b).

S1. The exact square-sum condition

Keep the original $H = L^2(X, r dx)$, original lower-bounded self-adjoint extension A , finite real cutoff b , and sharp projector $E_b = F_A((-\infty, b])$. Write $h_x = h_x^{(b)}$ for the actual evaluation representative constructed at DSP42–DSP44. Thus

$$(E_b f)(x) = (f, h_x)_r, \quad h_x \in E_b H, \quad e_A(x, x, b) = \|h_x\|_H^2. \quad (\text{S26-1})$$

The interior function $x \mapsto h_x$ is smooth as an H -valued function on every compact receiving patch; the boundary representative remains the one already proved there. Only the interior representatives, defined at every point of X , are required for the integrals below. The weighted L^2 space has the countable basis proved in E5. Let $\chi : X \rightarrow \mathbb{C}$ be measurable and finite almost everywhere for the original completed measure. Its multiplication operator is the densely defined closed map

$$D(M_\chi) = \{f \in H : \chi f \in H\}, \quad M_\chi f = \chi f. \quad (\text{S26-2})$$

Density follows by $f_N = 1_{\{|\chi| \leq N\}} f$: these inputs are in the domain and tend to f by dominated convergence. For closedness, if $f_k \rightarrow f$ and $\chi f_k \rightarrow g$ in H , choose a common subsequence converging almost everywhere for both limits by the summable squared-norm error argument. Then $g = \chi f$ almost everywhere. The weighted subsequence argument uses the original measure and is valid by exhaustion with finite weighted sets. Alternatively integrate the squared errors and use their summable series; Tonelli makes that series finite almost everywhere. This proves both assertions without assuming χ bounded.

Define the nonnegative, possibly infinite quantity

$$I_b(\chi) = \int_X |\chi(x)|^2 e_A(x, x, b) r(x) dx. \quad (\text{S26-3})$$

If $I_b(\chi) < \infty$, Cauchy–Schwarz in S26-1 gives

$$\int_X |\chi(x)(E_b f)(x)|^2 r(x) dx \leq I_b(\chi) \|f\|_H^2. \quad (\text{S26-4})$$

Hence the actual spectral range is contained in $D(M_\chi)$, and

$$R_\chi = M_\chi E_b : H \longrightarrow H \quad (\text{S26-5})$$

is everywhere defined and bounded, with its exact original multiplication action. It satisfies $R_\chi E_b = R_\chi$. For any countable orthonormal basis (v_j) , nonnegative Tonelli and complete Parseval give

$$\begin{aligned} \|R_\chi\|_2^2 &= \sum_j \int_X |\chi|^2 |(v_j, h_x)_r|^2 r dx \\ &= \int_X |\chi|^2 \sum_j |(v_j, h_x)_r|^2 r dx = I_b(\chi). \end{aligned} \quad (\text{S26-6})$$

Conversely, if the literal product $M_\chi E_b$ is everywhere defined and Hilbert–Schmidt, the same nonnegative equality gives $I_b(\chi) < \infty$ and S26-6. Thus S26-3 is necessary and sufficient for an everywhere-defined Hilbert–Schmidt product. It is not a necessary criterion for mere boundedness. When $b < a$, the original projector and h_x are zero, so every finite almost-everywhere χ gives the zero product and $I_b(\chi) = 0$. No atom at b is lost in any case.

S2. Full adjoint domain and the actual initial sandwich

Under the finite I_b condition, for $f \in D(M_{\overline{\chi}})$ and $u \in H$, all the following pairings are finite and

$$(R_\chi u, f)_r = (E_b u, M_{\overline{\chi}} f)_r = (u, E_b M_{\overline{\chi}} f)_r. \quad (\text{S26-7})$$

Thus $R_\chi^* f = E_b M_{\overline{\chi}} f$ on the actual dense multiplication domain. The first operator is bounded on all H ; the second initial product has domain exactly $D(M_{\overline{\chi}})$, since E_b is everywhere defined. Its graph is a restriction of that bounded graph. Given f in H , approximate it by the multiplication-domain truncations above. Their images under the bounded R_χ^* converge to its image at f . This proves the exact closure and full domains

$$\overline{E_b M_{\overline{\chi}}} = R_\chi^*, \quad D(R_\chi^*) = H, \quad R_\chi^* = E_b R_\chi^*. \quad (\text{S26-8})$$

This does not identify the initial unbounded product domain with H .

More generally let χ_1 and χ_2 have finite I_b , and retain a bounded complex Borel m satisfying the exact upper-spectral-support condition DSP45a, including its value at b . Set $C_m = m(A) = E_b m(A) E_b$, with its exact PVM essential norm $\|m\|_{F_{A,\infty}} = \|C_m\|$. Define

$$T_{12,m} = R_{\chi_1} C_m R_{\chi_2}^* : H \longrightarrow H. \quad (\text{S26-9})$$

Its two Hilbert–Schmidt factors and the complete earlier programme ideal proof give

$$T_{12,m} \in \mathcal{S}_1(H), \quad \|T_{12,m}\|_1 \leq \|m\|_{F_{A,\infty}} I_b(\chi_1)^{1/2} I_b(\chi_2)^{1/2}. \quad (\text{S26-10})$$

The initial literal sandwich $M_{\chi_1} m(A) M_{\overline{\chi_2}}$ has domain exactly $D(M_{\overline{\chi_2}})$. Indeed the right multiplication must be defined; for every such input its image under $m(A)$ is in $E_b H$, which S26-4 puts inside $D(M_{\chi_1})$. On this domain, S26-7 and the two compressions give exactly the action of S26-9. Dense truncation and its bounded extension therefore prove

$$\overline{M_{\chi_1} m(A) M_{\overline{\chi_2}}} = T_{12,m}, \quad D(T_{12,m}) = H. \quad (\text{S26-11})$$

For the original bounded χ_1, χ_2 , their multiplication domains are already H ; S26-11 is then the original everywhere-defined product itself. In particular the original $\chi_1 = \chi_2 = \chi$ compactly supported statement is preserved exactly, with the same weighted measure, operator and norm bound.

For an initial-domain input f , the original multiplier kernel gives the almost-everywhere action

$$(T_{12,m} f)(x) = \chi_1(x) \int_X e_m(x, y) \overline{\chi_2(y)} f(y) r(y) dy. \quad (\text{S26-12})$$

For every fixed x , its integral is absolutely convergent: $M_{\chi_2} f \in H$, and the row $e_m(x, \cdot)$ is the conjugate of the actual vector $m(A)^* h_x \in H$. Weighted Cauchy–Schwarz proves absolute integrability. S26-4 proves its whole output is L^2 after multiplication by χ_1 . For a general f in H , S26-12 is interpreted as the L^2 limit of these actual multiplication-domain truncations, whose outputs converge by boundedness of T . This identifies the exact integral receiver and its extension; it does not assert an absolutely convergent unbounded-weight integral for every arbitrary input.

S3. Exact complex diagonal trace

The full multiplier kernel from (DSP45) is

$$e_m(x, y) = (C_m h_y, h_x)_r. \quad (\text{S26-13})$$

It obeys $|e_m(x, x)| \leq \|C_m\| \|h_x\|^2$. Weighted Cauchy–Schwarz for the measure with density $\|h_x\|^2 r$ gives

$$\int_X |\chi_1 \overline{\chi_2} e_m(x, x)| r dx \leq \|C_m\| I_b(\chi_1)^{1/2} I_b(\chi_2)^{1/2} < \infty. \quad (\text{S26-14})$$

To prove the trace without an unproved integral-operator theorem, let P_N be the projection on the first N basis vectors. Hilbert–Schmidt tails give $R_{\chi_2} P_N \rightarrow R_{\chi_2}$ in that norm, so the factorization bound gives $R_{\chi_1} C_m P_N R_{\chi_2}^* \rightarrow T_{12,m}$ in trace norm. Its finite rank-one trace is

$$\sum_{j=1}^N (R_{\chi_1} C_m v_j, R_{\chi_2} v_j)_r = \int_X \chi_1 \overline{\chi_2} \sum_{j=1}^N (C_m v_j, h_x)_r (h_x, v_j)_r r dx. \quad (\text{S26-15})$$

Every finite integral is absolutely defined by the two L^2 factors. At a fixed x , the absolute coefficient sum is at most

$$\left(\sum_j |(C_m v_j, h_x)_r|^2 \right)^{1/2} \left(\sum_j |(h_x, v_j)_r|^2 \right)^{1/2} \leq \|C_m\| \|h_x\|^2. \quad (\text{S26-16})$$

Parseval shows its full complex sum is $(C_m h_x, h_x)_r$, in exactly that order for the linear-first pairing. S26-14 dominates the integrals. Dominated convergence and trace-norm continuity now prove

$$\text{Tr } T_{12,m} = \int_X \chi_1(x) \overline{\chi_2(x)} e_m(x, x) r(x) dx. \quad (\text{S26-17})$$

For m equal to the original sharp indicator and $\chi_1 = \chi_2 = \chi$, the operator is $R_\chi R_\chi^*$, positive, with trace and trace norm exactly $I_b(\chi)$. For two distinct weights or a complex multiplier the trace can cancel and the norm estimate S26-10 remains the asserted one. The vector-space $\{\chi : I_b(\chi) < \infty\}$ has the exact seminorm $I_b^{1/2}$, since it is the L^2 norm for the positive measure $e_A(x, x, b) r dx$. Its null space consists precisely of the weights vanishing for that measure, and S26-6 maps the quotient isometrically to its Hilbert–Schmidt multiplication-projector image. This identifies the space defined by the physical localization obstruction and its exact operator map.

S4. A singular global receiver with an exact trace

Keep the original half-line Dirichlet operator $A = -d^2/dx^2$ on $x > 0$, domain $H^2 \cap H_0^1$, positive constant original density r_0 and finite cutoff $\lambda \geq 0$. Keep both terms of DSP-E23:

$$e_A(x, x, \lambda) = \frac{1}{\pi r_0} \left(a - \frac{\sin(2ax)}{2x} \right), \quad a = \sqrt{\lambda}. \quad (\text{S26-18})$$

Take $\chi(x) = 1/x$ on the whole half-line. This weight is finite at every point of X , unbounded at its boundary and not in $L^2(X, r_0 dx)$. Its original multiplication domain is $\{f : \int_0^\infty |f(x)|^2 x^{-2} r_0 dx < \infty\}$, a proper dense domain: $1_{(0,1)}$ is in H and not in it. For $a > 0$ its exact spectral localization density is

$$q_\lambda(x) = |\chi(x)|^2 e_A(x, x, \lambda) r_0 = \frac{1}{\pi x^2} \left(a - \frac{\sin(2ax)}{2x} \right). \quad (\text{S26-19})$$

No density factor was discarded: the original kernel factor r_0^{-1} multiplies the actual measure $r_0 dx$, giving this exact comparison. Since $\sin t/t = \int_0^1 \cos(st) ds$, this density is nonnegative. The inequality $0 \leq 1 - \cos u \leq u^2/2$, proved by two integrations of its bounded second derivative, gives

$$0 \leq q_\lambda(x) \leq \frac{2a^3}{3\pi}, \quad \lim_{x \downarrow 0} q_\lambda(x) = \frac{2a^3}{3\pi}. \quad (\text{S26-20})$$

The limit follows by dominated convergence in the same cosine integral after division by t^2 ; its integral of $s^2/2$ is $1/6$. At infinity the bounded sine in S26-19 gives integrable decay, with $q_\lambda(x)$ asymptotic to $a/(\pi x^2)$. Thus its integral is finite despite the unbounded weight and infinite global spectral rank.

Its exact value follows from the original Fourier Plancherel factor. For $f = 1_{(0,1)}$, $\hat{f}(t) = (1 - e^{-it})/(it)$ for $t \neq 0$ and one at $t = 0$. Its L^1 and L^2 Fourier transforms agree by P001 FL4, and FL5 gives

$$2 \int_{\mathbb{R}} \frac{1 - \cos t}{t^2} dt = \int_{\mathbb{R}} |\hat{f}(t)|^2 dt = 2\pi. \quad (\text{S26-21})$$

Evenness therefore gives the positive half integral $\pi/2$. For $0 < s \leq 1$, the substitution $u = st$ retains the full factor s and gives $\int_0^\infty (1 - \cos(st)) t^{-2} dt = s\pi/2$; at $s = 0$ the integral is zero. Nonnegative Tonelli now proves

$$\int_0^\infty \frac{1 - \sin t/t}{t^2} dt = \int_0^1 \int_0^\infty \frac{1 - \cos(st)}{t^2} dt ds = \frac{\pi}{4}. \quad (\text{S26-22})$$

In S26-19 put $t = 2ax$. Retaining $dx/x^2 = 2a dt/t^2$ and the factor a inside the parenthesis gives

$$I_\lambda(1/x) = \frac{2a^2}{\pi} \int_0^\infty \frac{1 - \sin t/t}{t^2} dt = \frac{a^2}{2} = \frac{\lambda}{2}. \quad (\text{S26-23})$$

Hence the literal $M_{1/x} E_\lambda$ is an everywhere-defined Hilbert–Schmidt map with squared norm $\lambda/2$. The positive trace-class closure of the initial original sandwich $M_{1/x} E_\lambda M_{1/x}$ has trace and trace norm $\lambda/2$. Its initial domain remains the proper dense domain above; its closure is the whole bounded operator specified by S26-11. The global projector itself still has infinite rank and infinite trace for $\lambda > 0$, as E6

proves. For $\lambda = 0$, the ball projector and all kernels are zero, so $I_\lambda(1/x) = 0$ and both products and the closure are zero. A negative cutoff also has zero projector for this nonnegative A . Every exceptional case is retained.

For a finite physical window $L > 0$, the same singular trace can also be evaluated exactly. Put $t = 2aL$ and define $\text{Si}(t) = \int_0^t \sin(u)/u \, du$, with the integrand's original value one at $u = 0$. Differentiation verifies

$$\int_0^t \frac{1 - \sin u/u}{u^2} \, du = \frac{\text{Si}(t)}{2} + \frac{\cos t - 2}{2t} + \frac{\sin t}{2t^2}. \quad (\text{S26-24})$$

The derivative is exactly the integrand; at zero all reciprocal terms cancel and its limit is zero, either from their power series or S26-20. To verify this endpoint with an explicit remainder, denote the right side of (S26-24) by $G(t)$. Applying the finite Taylor formula (OC22) to the original sine and cosine functions, whose derivatives have absolute value at most one, gives for $0 < t \leq 1$

$$\left| \sin t - t + \frac{t^3}{6} \right| \leq \frac{t^5}{120}, \quad \left| \cos t - 1 + \frac{t^2}{2} \right| \leq \frac{t^4}{24}, \quad |\text{Si}(t) - t| \leq \int_0^t \frac{u^2}{6} \, du = \frac{t^3}{18}.$$

For the last bound, the same Taylor formula gives $|\sin u - u| \leq u^3/6$, before division by u ; the value at zero is its continuous limit. Substitute these three remainders into the full reciprocal expression for G . Its reciprocal leading terms cancel exactly, its remaining leading term is $t/6$, and every remainder is retained in the bound

$$\left| G(t) - \frac{t}{6} \right| \leq t^3 \left(\frac{1}{36} + \frac{1}{48} + \frac{1}{240} \right) = \frac{19t^3}{360}.$$

Thus $G(t) \rightarrow 0$ at zero, proving the integration constant in (S26-24). Therefore

$$\int_0^L q_\lambda(x) \, dx = \frac{a^2}{\pi} \left(\text{Si}(2aL) + \frac{\cos(2aL) - 2}{2aL} + \frac{\sin(2aL)}{(2aL)^2} \right). \quad (\text{S26-25})$$

Its continuous value is zero at $L = 0$ or $a = 0$. This is an additional singular-window formula, not a replacement for the distinct original bounded-window formula DSP-E23a. Its derivative is $q_\lambda \geq 0$ and its limit at infinity is the already proved $\lambda/2$, so it obeys that exact upper bound. The figure below displays the exact multiplication-domain and closure maps in S26-1–S26-17, together with numerical samples of the proved one-dimensional formulas S26-19–S26-25. It uses the original density $r_0 = 2$ and cutoffs $\lambda = 1$ and 4. The continuous boundary values and the exact whole-half-line trace $\lambda/2$ are labelled explicitly.

Figure S26. For the original operator $A = -d^2/dx^2$ with domain $H^2 \cap H_0^1$, the two kernel terms give the full weighted localization density q_λ of (S26-19). The left panel retains its exact boundary limit $2\lambda^{3/2}/(3\pi)$; the right panel samples (S26-25) and marks its proved limit $\lambda/2$. The upper maps keep the proper dense initial multiplication domain and its bounded graph closure. These are numerical sections of the proved half-line formulas, rather than a claim about finite global spectral rank. Reproducible Python source · Native vector output.

A singular physical weight has an exact spectral localization space

Original $A = -d^2/dx^2$, $x > 0$; domain $H^2 \cap H_0^1$; density $r_0 = 2$; $a = \sqrt{\lambda}$

$$e(x, x, \lambda) = [a - \sin(2ax)/(2x)] / (2\pi); \quad q\lambda(x) = [a - \sin(2ax)/(2x)] / (\pi x^2)$$

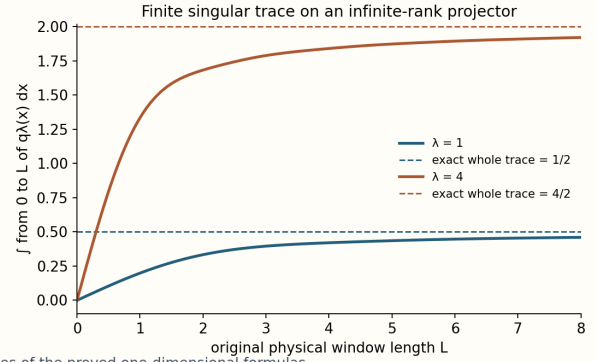
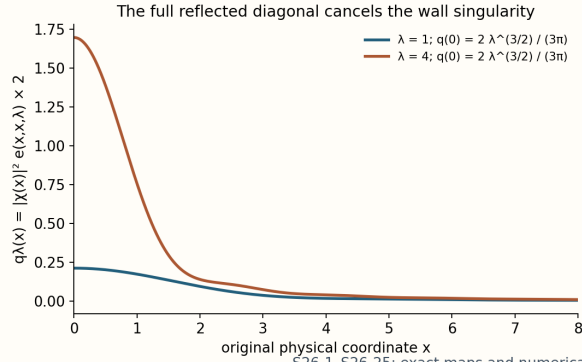
Original $H = L^2((0, \infty), 2 dx)$
 $\chi(x) = 1/x$; $E\lambda$ retains the closed cutoff

$R\chi = M\chi E\lambda$ is defined on all H
 Hilbert-Schmidt: $\|R\chi\|_2^2 = \lambda/2$

Initial $M\chi E\lambda M\chi$; domain $D(M\chi)$
 $D(M\chi) = \{f : \int |f(x)|^2 2 dx / x^2 < \infty\}$

exact graph closure $\rightarrow$

Its closure is $R\chi R\chi^*$ on all H
 Positive trace and trace norm $= \lambda/2$



S26-1-S26-25: exact maps and numerical samples of the proved one-dimensional formulas.
 The original kernel factor $1/2$ multiplies the original measure $2 dx$. $\lambda = 0$ gives zero; all wall limits are retained.

Figure 4: The original half-line Dirichlet operator with weight $1/x$: exact multiplication-domain and closure maps, reflected spectral density and finite singular traces

S5. Propagation at the original scope

DSP-E14–DSP-E17 remain valid as their bounded compactly supported specialization. S26-3 removes both restrictions exactly when its weighted diagonal integral is finite, and S26-8–S26-12 supplies the required unbounded adjoint/domain/closure maps. DSP-E15 still gives a sufficient condition for every L^2 weight supported in an allowed compact set, since the original diagonal estimate bounds I_b by its full local constant times the squared physical L^2 norm; boundedness of such a weight is unnecessary. S26-10 and S26-17 also strengthen the original same-weight statement to two different weights with every conjugation and original multiplication order retained. DSP-E18–DSP-E23a supply the unchanged reflected/density kernel used in the global $1/x$ example. Both Sobolev endpoint obstructions and the global infinite-rank conclusion remain valid. The new space and its operator map receive precisely the obstruction to global trace without asserting compact resolvent or global finite spectral rank.

For the following comparison with E5, return to Section 10's original arbitrary lower-bounded extension A , density r , lower bound a , shift ρ , dimension n and finite real cutoff b . Here a denotes that original lower bound; the half-line calculation in S4 used E6's distinct local notation $a = \sqrt{\lambda}$.

Here is the full comparison with the original local bound. Let K be a compact union of the allowed receiving patches from E5, and let $\chi \in L^2(X, r dx)$ vanish almost everywhere outside K . If $b < a$, its projector is zero, so every localization formula is zero. If $b \geq \max(1, a)$, keep the exact constant and parameter in (DSP43):

$$I_b(\chi) \leq C_{0,K}^2 (b + \rho)^{n/2} \int_X |\chi(x)|^2 r(x) dx < \infty.$$

The factor $C_{0,K}^2$ is the actual square of the original evaluation constant in (DSP43), at multiindex zero. The original bound (DSP-E15) keeps its stated $C_{\text{supp } \chi}$ for a diagonal-bound constant; the present comparison displays the square from which that bound is obtained. No boundedness of χ enters this estimate. For any remaining finite $b \geq a$, put $B_b = \max(1, b)$. The original projection-valued calculus gives $E_b E_{B_b} = E_b$. For every $f \in H$, evaluating this equality gives

$$(f, h_x^{(b)})_r = (E_b f, h_x^{(B_b)})_r = (f, E_b h_x^{(B_b)})_r, \quad h_x^{(b)} = E_b h_x^{(B_b)}.$$

The contraction bound therefore gives $e_A(x, x, b) \leq e_A(x, x, B_b)$, and the same displayed estimate holds with its full factor $(B_b + \rho)^{n/2}$. Thus every such physical L^2 weight, including an unbounded one, has the literal everywhere-defined Hilbert–Schmidt product (S26-5). Equations (S26-8)–(S26-11) prove the exact domains and graph closures of its adjoint restriction and sandwich. With the original bounded weights those domains were already H , so the closures equal the literal products in (DSP-E14)–(DSP-E17). With two different weights of finite I_b , the same proof gives (S26-10) and (S26-17), retaining the right-hand conjugation, the original weighted measure, and the full essential spectral norm. This proves the stated propagation to E5 at every finite cutoff.

For the half-line comparison with E6, retain its original $A = -d^2/dx^2$, domain $H^2 \cap H_0^1$, density $r_0 > 0$ and notation $a = \sqrt{\lambda}$ for $\lambda \geq 0$. The propagation to E6 uses its original density comparison (DSP-E22), followed by both terms in (DSP-E23). Multiplication by $r_0 dx$ and the squared singular weight gives exactly (S26-19); the Fourier calculation (S26-21)–(S26-23) then gives the finite value $\lambda/2$. The original global projector remains of infinite rank when $\lambda > 0$. Indeed its diagonal tends to $a/(\pi r_0) > 0$, as E6 proves. Choose $R > 0$ so that $e_A(x, x, \lambda) r_0 \geq a/(2\pi)$ for every $x \geq R$. For $L > R$, the original bounded localization $\eta_L = 1_{(R,L)}$ is an E5 weight, and its full trace is

$$\begin{aligned}
\text{Tr}(M_{\eta_L} E_\lambda M_{\eta_L}) &= \int_R^L e_A(x, x, \lambda) r_0 dx \\
&= \frac{a(L-R)}{\pi} - \frac{\text{Si}(2aL) - \text{Si}(2aR)}{2\pi} \geq \frac{a(L-R)}{2\pi}.
\end{aligned}$$

The integral comparison proves the last inequality without removing either sine-integral boundary contribution from the exact middle expression. If E_λ were trace class, the earlier ideal inequality (T9), with both multiplication norms at most one, would bound every one of these traces by $\|E_\lambda\|_1$, contradicting their growth. A finite-rank projection has finite trace equal to its rank by (T12)–(T13), so the rank is infinite as well. At $\lambda \leq 0$, the original projector is zero and this positive-cutoff argument is not applied. The finite singular trace and the infinite global trace consequently concern their exact different operators, related by the multiplication and closure maps already proved above. The bounded window (DSP-E23a) and the singular window (S26-25) both retain their complete boundary terms and their continuous zero-parameter values.

This model also verifies why the criterion (S26-3) was asserted for Hilbert–Schmidt localization rather than for boundedness alone. With $\chi \equiv 1$ and $\lambda > 0$, the literal product is the original bounded projection $M_1 E_\lambda = E_\lambda$, while $I_\lambda(1) = \int_0^\infty e_A(x, x, \lambda) r_0 dx = \infty$ by the same positive diagonal tail. Thus finite I_b is unnecessary for boundedness, and its necessity for the Hilbert–Schmidt product remains exactly the equality (S26-6).